ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

DEPARTMENT OF PHYSICS AND ASTRONOMY "A. RIGHI"

SECOND CYCLE DEGREE

PHYSICS

# Clinical Outcome Analysis of Ultrasound-Guided Stereotactic Body Radiotherapy (SBRT)

Supervisor
Prof. Claudia Testa

Co-supervisor
Dott. Lidia Strigari

Defended by
Yamna Siddiqui

Graduation Session / Month/ Year

Academic Year 2024/2025

*The important thing is not to stop questioning. Curiosity has its own reason for existing. One cannot help but be in awe when one contemplates the mysteries of eternity, of life, of the marvellous structure of reality. It is enough if one tries to comprehend only a little of this mystery every day.*

Albert Einstein

# Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

# Contents

**Abstract** **i**

**List of Figures** **v**

**List of Tables** **vii**

**1 Introduction** **1**
1.1 Clinical Background . . . . . 1
1.2 Geometric Accuracy Requirements in Prostate SBRT . . . . . 1
1.3 Inter- and Intrafraction Prostate Motion . . . . . 2
1.4 Existing Image-Guided Radiotherapy Techniques . . . . . 2
1.5 Ultrasound-Guided Radiotherapy for Prostate SBRT . . . . . 3
1.6 Summary of Existing Evidence on Ultrasound-Guided Prostate Radiotherapy 4
1.7 Research Gap and Aim of the Study . . . . . 4

**2 Methodology** **5**
2.1 Study Design . . . . . 5
2.2 Literature Search Strategy . . . . . 6
2.3 Study Selection Criteria . . . . . 6
2.4 Data Extraction . . . . . 7
2.5 Quantitative Analysis . . . . . 7
2.5.1 Stochastic Motion Modeling . . . . . 7
2.5.2 Statistical Outcome Analysis . . . . . 8
2.6 Limitations of the Method . . . . . 8

**3 Literature Review** **9**
3.1 Overview of Prostate Motion and Setup Uncertainties . . . . . 9
3.2 Image Guidance Techniques for Prostate Radiotherapy . . . . . 10
3.2.1 Cone-Beam Computed Tomography (CBCT) . . . . . 10
3.2.2 Fiducial Marker-Based Localization . . . . . 10
3.2.3 Magnetic Resonance Image-Guided Radiotherapy (MRIgRT) . . 10
3.2.4 Electromagnetic (EM) Tracking . . . . . 11
3.2.5 Ultrasound Guidance . . . . . 11
3.3 Ultrasound-Based Guidance: Principles and Technologies . . . . . 11
3.4 Setup Accuracy and Target Localization Using Ultrasound . . . . . 12
3.5 Intrafraction Motion Monitoring and Validation . . . . . 13
3.6 Clinical Outcome Evidence of Ultrasound-Guided Prostate Radiotherapy 16
3.7 Summary and Identified Knowledge Gaps . . . . . 17

**4 Quantitative and Clinical Comparison of Radiotherapy Techniques 18**
4.1 Comparison of PTV Margin Strategies . . . . . 18
4.1.1 The Evolution from Static to Dynamic Margin Paradigms . . . . 19
4.1.2 Mathematical Modeling of Intrafraction Motion: The Random Walk . . . . . 22
4.1.3 Stochastic Modeling of Intrafraction Prostate Motion . . . . . 23
4.1.4 Duration-Dependent Margins and the "Random Walk" . . . . . 24
4.2 Imaging Techniques and Setup Accuracy . . . . . 25
4.2.1 Correlation Between Modality and Localization Precision . . . . 26
4.2.2 The Impact of Operator Experience and the Learning Curve . . . 27
4.2.3 Technical Considerations in Ultrasound-Based Guidance . . . . 28
4.3 Machine-Based Comparison and Technology Impact . . . . . 29
4.3.1 Technological Influence on Targeting and Efficiency . . . . . 29
4.3.2 Comparative Analysis of Treatment Platforms . . . . . 30
4.3.3 Clinical Advantages of Fiducial-Free Real-Time Tracking . . . . . 32
4.3.4 System-Specific Advantages and Constraints . . . . . 33
4.4 Comparison of Adaptive and Non-Adaptive Radiotherapy . . . . . 34
4.4.1 Physiological Drivers of Prostate Displacement . . . . . 35
4.4.2 Technological Drivers of Adaptation . . . . . 35
4.4.3 Comparative Analysis of Adaptive Strategies . . . . . 36
4.4.4 Impact on Clinical Efficiency and Toxicity . . . . . 38
4.4.5 Intrafraction Intervention Rate Analysis . . . . . 38
4.5 Dosimetric Outcomes . . . . . 40
4.5.1 Dose Conformity and Gradient Steepness . . . . . 41
4.5.2 Impact of Real-Time Guidance on OAR Sparing . . . . . 41
4.5.3 Comparative Dosimetric Data Table . . . . . 42
4.5.4 Dosimetric Stability and Motion Thresholds . . . . . 44
4.6 Comparison of Clinical Outcomes and Toxicity . . . . . 44
4.6.1 Weighted Clinical Outcome Synthesis . . . . . 45
4.6.2 Tumor Control and PSA Response . . . . . 46
4.6.3 Comparative Toxicity Profiles . . . . . 47
4.6.4 The Impact of Real-Time Adaptation on Side Effects . . . . . 49
4.6.5 Regression Analysis: PTV Margin Size and Gastrointestinal Toxicity 50
4.6.6 Model-Based Clinical Outcome Projection: VMAT vs SS-IMRT (Higuchi et al. 2022) . . . . . 52
4.7 Synthesis of Technical Trade-offs in Prostate Guidance . . . . . 53
4.8 Comparative Analysis Matrix . . . . . 54
4.9 Synthesis of Key Findings . . . . . 56
4.9.1 Concluding Remarks . . . . . 57

**5 Results, Conclusion and Future Development** **61**
5.1 Results . . . . . 61
5.1.1 Main Numerical Findings and Trends . . . . . 61
5.1.2 Model-Based Clinical Outcome Projection: VMAT vs SS-IMRT . . 63
5.1.3 Regression Robustness and Sensitivity Analysis . . . . . 65
5.1.4 Comparative Performance of Guidance Modalities . . . . . 66
5.1.5 Clinical and Dosimetric Impact . . . . . 68
5.1.6 Consistency Across the Dataset . . . . . 71
5.2 Cost-Effectiveness and Clinical Feasibility Analysis . . . . . 71
5.3 Simulation-Derived Margin Requirements . . . . . 74
5.4 Limitations of the Study . . . . . 76
5.5 Critical Evaluation of the Evidence . . . . . 77
5.6 Conclusion . . . . . 79
5.6.1 Final Clinical Statements . . . . . 79
5.6.2 Balancing Technological Advancements . . . . . 81
5.6.3 Closing Summary . . . . . 81

**Acknowledgement** **83**

**6 Acknowledgement** **83**

**Appendix** **84**

**References** **85**

# List of Figures


1 Conceptual overview of the clinical challenge of intra-fraction prostate motion and the role of real-time transperineal ultrasound (TP-US) monitoring to improve targeting accuracy and enable smaller treatment margins. . . . . 5

2 Comparison of PTV margins across different imaging techniques. . . . . 22

3 Probability of prostate displacement exceeding clinical thresholds of 2 mm, 3 mm, and 5 mm as a function of treatment duration, estimated via Monte Carlo random walk simulation ($n = 5000$, $\sigma = 0.18$ mm/s, based on Ballhausen et al. 2015). Vertical dashed lines indicate typical VMAT (≈120 s) and SS-IMRT (≈600 s) delivery durations. The shaded region highlights the typical VMAT delivery window. . . . . 24

4 Left: Intrafraction correction rates across guidance strategies. Green bars represent adaptive real-time TPUS-based workflows with defined motion thresholds. Red bars represent non-adaptive CBCT-only workflows, where motion occurs but is not detected or corrected. Right: Linear regression of motion threshold versus intervention rate across adaptive studies ($R^2 = 0.88$). Marker size is proportional to patient cohort size. . 40

5 Linear regression analysis of PTV margin size versus late Grade 2+ gastrointestinal toxicity across five published studies. The regression line (blue) is shown with 95% confidence interval (shaded region). Pearson $R = 0.873$, $R^2 = 0.762$, $p = 0.053$, slope $= 1.973$% per mm. . . . . 51

6 Schematic representation of intrafraction motion management pathways and their impact on treatment accuracy, toxicity reduction, and clinical outcomes in prostate SBRT. . . . . 59

7 Comparison of setup uncertainty and localization precision across selected prostate IGRT studies. Mean displacement is shown with $\pm 1$ standard deviation. The shaded region (0–3 mm) represents the clinically acceptable precision range for high-accuracy SBRT delivery. . . . . 62

8 Four-panel model-based comparison of VMAT and SS-IMRT. Top left: treatment delivery time. Top right: clinical versus effective PTV margin (including motion contribution). Bottom left: model-predicted Grade ≥2 GI toxicity derived from the regression equation $T = 1.973 \times M - 3.027$. Bottom right: mechanistic pathway from delivery time to clinical toxicity. All geometric inputs are extracted from Higuchi et al. (2022). This constitutes a model-extrapolated prediction rather than a directly observed clinical endpoint. . . . . 65

9 Impact of guidance technology on PTV margin selection across analyzed studies. Error bars represent estimated variability based on reported ranges. . . . . 68
10 Multi-criteria comparison of IGRT modalities across five clinically relevant performance axes. Scores are based on literature-derived qualitative and quantitative assessments. . . . . 70
11 Four-panel cost-effectiveness comparison of IGRT modalities. Capital cost and treatment time data are based on institutional reports and published system specifications. PTV margins are derived from Table 19. Toxicity rates are sourced from Fonteyne et al. (2011), Lee et al. (2021), and Takai et al. (2024). Error bars represent reported ranges across studies. 73

# List of Tables


1 Summary of reported setup accuracy and margin recommendations for ultrasound-guided prostate radiotherapy. . . . . 13
2 Representative studies on intra-fraction prostate motion and monitoring strategies, summarizing methods, cohorts, key findings, and limitations. 14
3 Summary of reported intrafraction prostate motion and correction strategies using ultrasound guidance. . . . . 16
4 Summary of reported clinical outcomes for ultrasound-guided prostate radiotherapy. . . . . 17
5 Comparative Analysis of PTV Margin Strategies and Associated Modalities 21
6 Comparison of Imaging Modalities and Setup Accuracy Metrics . . . . . 27
7 Comparative Performance of Radiotherapy Delivery Platforms . . . . . 31
8 Comparison of Adaptive and Non-Adaptive Radiotherapy Metrics . . . 37
9 Intrafraction correction rates across selected prostate radiotherapy studies. 39
10 Comparison of Dosimetric Outcomes by Technique and Guidance . . . 43
11 Data used in the weighted clinical outcome synthesis, including confirmed and estimated sample sizes. . . . . 45
12 Comparison of Clinical Outcomes and Toxicity Across Selected Studies . 48
13 Data points used in the regression analysis of PTV margin size versus late Grade 2+ gastrointestinal toxicity. . . . . 50
14 Comparative Synthesis of Prostate Guidance and Tracking Modalities . . 53
15 Comparative Matrix of Radiotherapy Techniques . . . . . 55
16 Summary of setup accuracy and localization precision across selected prostate IGRT studies . . . . . 62
17 Clinical and dosimetric parameters extracted from Higuchi et al. (2022) for model-based outcome projection. . . . . 63
18 Leave-one-out sensitivity analysis of regression model . . . . . 66
19 Derivation of PTV margin values across imaging and motion management strategies . . . . . 67
20 Multi-criteria evaluation of IGRT modalities based on literature-derived performance indicators . . . . . 69
21 Comparison of cost and operational characteristics of IGRT modalities. . 73
22 Simulation-derived margin requirements for motion coverage . . . . . 75

# 1 Introduction

## 1.1 Clinical Background

A significant and public health concern nowadays, particularly in the aging male population, is prostate cancer[1, 2]. Prostate cancer is becoming one of the most diagnosed malignancies among men worldwide[3]. Because of its slow-growing nature and high long-term survival rates, we must choose a treatment that balances the quality of life with effective tumor control. The main goal in prostate cancer is to minimize treatment-related toxicity and also ensure clinical efficacy[1, 4].

For curative treatments, radiotherapy is a well-known treatment option for patients with cancer, especially prostate cancer, and it is commonly used as a substitute for radical prostatectomy [2]. Advancements in external beam radiotherapy have improved overall dose conformity and reduced sparing of surrounding organs at risk, such as the bladder and rectum [4, 1]. As seen in Clinical trials demonstrating a clear dose–response relationship in prostate cancer, higher radiation doses can improve biochemical control without a significant increase in adverse effects when delivered through modern techniques [4, 1].

In recent years, stereotactic body radiotherapy (SBRT) has emerged as a promising treatment approach for prostate cancer [2, 4]. SBRT delivers high doses of radiation per fraction over a small number of treatment sessions [4]. Compared with the conventional fractionation schedule, SBRT typically requires only five fractions. Overall, SBRT offers several advantages, including increased patient convenience, reduced treatment time, and a lower toxicity profile [4, 1].

Intrafraction motion is a key challenge in prostate SBRT because it affects how precisely the treatment can be delivered. This makes it important to use advanced methods to manage motion.

## 1.2 Geometric Accuracy Requirements in Prostate SBRT

To achieve greater clinical benefits from stereotactic body radiotherapy, it should be accompanied by stringent geometric accuracy requirements. While planning the treatment plan for prostate cancer with SBRT, volume margins are employed very small, typically in a few millimeters, to protect the surrounding organs at risk. [5, 6]. On the other hand, SBRT is characterized by steep dose gradients, meaning the radiation dose decreases rapidly outside the target. As a result, a little error in setup or target displacement can cause a great underdose to the tumor or overdose to surrounding organs at risk, such as the bladder and rectum. [6, 4].

During SBRT, the delivered dose per fraction is high. Geometrical uncertainties can amplify clinical and biological effects compared to conventional radiotherapy. Much

research has shown that even millimeter-level inaccuracies may compromise tumor control or increase toxicity. Extreme precision is fundamental throughout the therapy. This is required for the safe and effective implementation of SBRT in prostate cancer [6, 5, 4].

## 1.3 Inter- and Intrafraction Prostate Motion

One of the main obstacles in prostate SBRT radiotherapy is the motion of the prostate because of its relation to surrounding anatomical structures. We can categorize prostate motion into two components: inter-fraction and intra-fraction; both contribute to geometric uncertainties in treatment delivery [6, 4].

Inter-fraction prostate motion is the motion of the prostate that occurs between the fraction days. Inter-fraction is mainly affected by changes in rectal volume, patient setup reproducibility, and bladder filling. During inter-fraction motion, several millimeters of displacement are observed, indicating that the prostate motion at the time of planning treatment does not remain the same over the course of the fraction days [2, 4].

When the prostate moves during radiation delivery, this is called intrafraction motion. The movement can occur on its own or be caused by a sudden change. Many studies have shown that the longer the treatment time, the more the prostate drifts [6, 4]. During SBRT, intrafraction motion needs extreme precision because of the high doses per fraction. Any error may amplify biological effects [5].

Because of these two types of prostate motion, the dose coverage can be reduced, and surrounding organs at risk may receive unwanted radiation exposure. This may also cause high negative toxicity rates in the genitourinary and gastrointestinal systems, which can badly affect the tumor[4, 1]. It is crucial to control prostate motion during SBRT. Some image-guided radiotherapies address this by locating and correcting the prostate's motion during treatment. This improves precision and geometric accuracy [1, 2].

## 1.4 Existing Image-Guided Radiotherapy Techniques

To reduce uncertainties in geometric accuracy and localization precision during prostate motion, many image-guided radiotherapy techniques have been developed. Cone beam computed tomography (CBCT) is the most commonly used approach. CBCT provides detailed 3D anatomical information before treatment[4, 2]. It also verifies prostate motion, which can be compared to the planning computed tomography scan. However, CBCT delivers additional imaging dose and is valid only for pretreatment imaging. It provides only limited information about intrafraction motion[2, 4].

Another widely used technique is the implantation of fiducial markers within the prostate, which we can track using X-ray imaging systems. This technique is beneficial for improving localization, but it involves substantial exposure to radiation. Not only

this, but it relies on intermittent imaging, which limits its ability to track the prostate during the treatment [4]. Electromagnetic tracking systems provide real-time position information without ionizing radiation, but they need the implantation of transponders, which may increase the cost and workflow complexities.[2].

Another advanced IGRT solution is magnetic resonance image-guided radiotherapy (MRIgRT). It provides superior soft tissue contrast and potential for real-time imaging. Although these techniques have advantages, they are not widely used. The systems are resource-intensive and expensive, which limits clinical accessibility[1].

Altogether, the IGRT techniques have remarkably improved the localization of the prostate during treatment. But it has many drawbacks, including invasiveness, additional imaging dose, insufficient information for real-time monitoring during treatment, and high costs. This arises from a need for alternative image-guidance solutions that are accurate, non-invasive, and provide real-time prostate monitoring during SBRT [4, 2, 1].

## 1.5 Ultrasound-Guided Radiotherapy for Prostate SBRT

One promising IGRT technique is ultrasound-guided radiotherapy for prostate cancer treatment using SBRT. Ultrasound-guided imaging visualizes soft tissue structures without invasiveness. It can be integrated into standard linear accelerator workflow. This technique does not increase imaging dose. [4, 2].

Among many ultrasound-based techniques, transperineal ultrasound (TPUS) is the most popular for prostate radiotherapy. To set up TPUS-guided systems, the ultrasound probe is positioned in contact with the perineum. This position allows for localization of the prostate and detecting small changes like bladder filling or abdominal motion during treatment[1]. This arrangement improves image consistency and reduces probe-induced prostate displacement, especially compared with transabdominal ultrasound techniques [4].

TPUS-guided radiotherapy offers a great advantage in monitoring the real-time prostate motion during treatment delivery. This advantage enables the detection of intrafractional prostate motion and allows timely correction when predefined motion thresholds are exceeded [1]. Also, in TPUS, there is no need for implanted markers or transponders; it is safe, non-invasive, reduces the imaging dose, reduces patients' discomfort, and simplifies clinical workflows [2].

In summary, the combination of no ionizing radiation, real-time motion monitoring, and non-invasive methods makes TPUS-guided systems an attractive solution for geometric uncertainties in prostate SBRT[4, 2, 1].

## 1.6 Summary of Existing Evidence on Ultrasound-Guided Prostate Radiotherapy

An extensive literature review indicates that the use of US guidance in prostate radiotherapy addresses multiple aspects of clinical outcomes, motion management, and treatment accuracy. Instead of examining individual studies, we would rather focus on existing evidence that can be categorized by its primary clinical and technical objectives [1].

Many studies have shown that US-guidance has significantly improved treatment planning and setup accuracy [4, 2]. 3D-TPUS imaging provides prostate localization and enables accurate alignment relative to the planning computed tomography scan [1, 4]. In this way, they established image-guidance methods that reduce setup uncertainties and tighter planning margins [4, 5].

Several studies also show the capability of US-based systems to monitor the prostate motion during radiation delivery [6, 4, 1]. Evidence shows that in clinical and experimental trials, TPUS can detect prostate motion during intrafraction with decent temporal resolution, thereby identifying clinically relevant displacements [6, 4]. Ultrasound monitoring shows that the prostate can move during treatment, making real-time motion management especially important for SBRT [6, 1].

In addition to the experimental trials, clinical outcome studies indicate that ultrasound-guided prostate radiotherapy provides effective tumor control with acceptable genitourinary and gastrointestinal toxicity compared with other image-guided radiotherapy methods. [1, 4, 2].

## 1.7 Research Gap and Aim of the Study

Many studies have looked at how ultrasound guidance improves setup accuracy and motion tracking in prostate radiotherapy. However, fewer studies have looked at how these improvements actually affect patient outcomes. There is also a lack of direct comparisons between ultrasound-guided SBRT and other image-guided techniques. This gap makes it difficult to fully understand the clinical value of ultrasound guidance in prostate SBRT. This study aims to fill that gap. The primary research question is: Does real-time transperineal ultrasound guidance enable a measurable reduction in PTV margins and treatment-related toxicity in prostate SBRT, compared to non-adaptive image-guided techniques? To answer this, three specific questions are explored:

How does the choice of image guidance technique affect PTV margin size and clinical toxicity outcomes? How does intrafraction prostate motion change over treatment time, and what does this mean for margin calculation in VMAT versus SS-IMRT delivery? Does real-time ultrasound guidance achieve comparable geometric accuracy to invasive tracking methods, while being more practical to use clinically?

The hypothesis of this study is that real-time transperineal ultrasound guidance

allows margins to be safely reduced from the conventional 7-10mm down to 2-3mm. It is further expected that this margin reduction leads to a measurable decrease in Grade 2+ gastrointestinal and genitourinary toxicity. These expectations are tested through stochastic motion modeling and a structured synthesis of clinical outcome data from published studies. As shown in Figure 1, unmanaged intrafraction prostate motion directly threatens treatment accuracy. This makes real-time monitoring not just helpful, but necessary for safe and precise prostate SBRT.

This study aims to determine whether real-time ultrasound guidance can shift SBRT from relying on treatment margins to a more precision-focused approach.

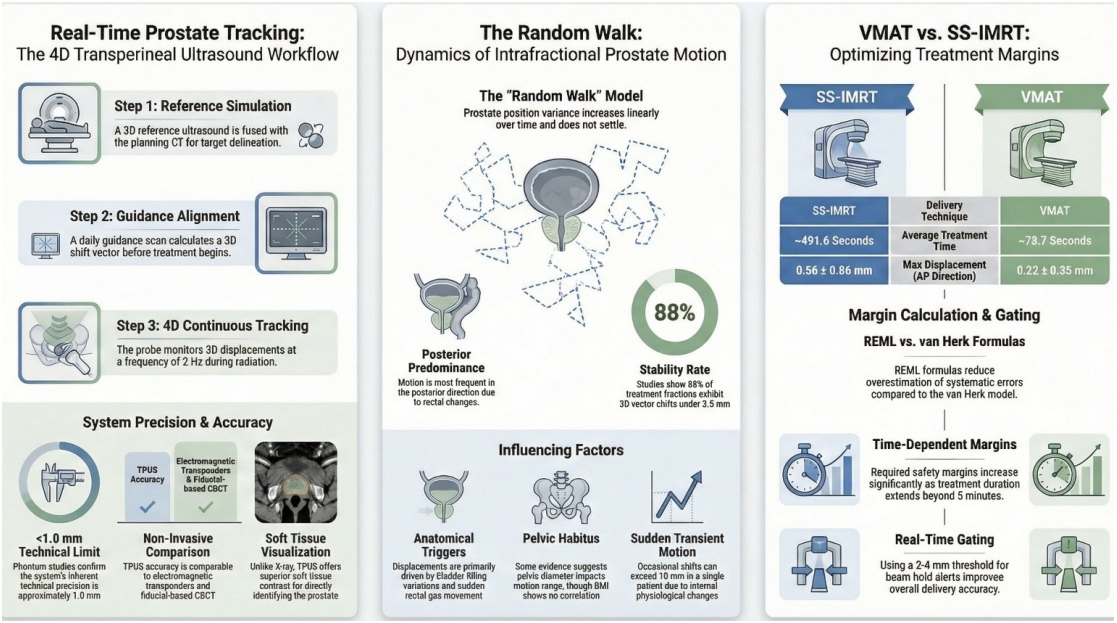


**Figure 1:** Conceptual overview of the clinical challenge of intra-fraction prostate motion and the role of real-time transperineal ultrasound (TP-US) monitoring to improve targeting accuracy and enable smaller treatment margins.

As shown in Figure 1, intra-fraction prostate motion can reduce target coverage, motivating real-time monitoring using TP-US.

# 2 Methodology

## 2.1 Study Design

This study is a systematic literature review combined with quantitative analysis. It was designed to evaluate the clinical and dosimetric outcomes of ultrasound-guided

stereotactic body radiotherapy for prostate cancer. The analysis compares ultrasound-based guidance with other image-guided radiotherapy techniques, focusing on planning target volume margins, intrafraction motion, and treatment-related toxicity.

## 2.2 Literature Search Strategy

A systematic search of the PubMed database was conducted to identify relevant studies. The following search terms were used in combination:

- ”prostate SBRT ultrasound guidance”
- ”transperineal ultrasound radiotherapy”
- ”intrafraction prostate motion SBRT”
- ”image-guided radiotherapy prostate cancer”
- ”stereotactic body radiotherapy prostate outcomes”

No restriction was placed on publication year in order to capture the full historical development of ultrasound-guided radiotherapy techniques. All searches were conducted in English.

## 2.3 Study Selection Criteria

Studies were included if they met the following criteria:

- The study focused specifically on prostate cancer radiotherapy
- The study reported quantitative data on at least one of the following: setup accuracy, PTV margin size, intrafraction motion magnitude, or clinical toxicity outcomes
- The study involved stereotactic body radiotherapy or hypofractionated radiotherapy delivery
- The study used an identifiable image guidance technique

Studies were excluded if they focused on non-prostate cancers, did not report quantitative outcomes, or were review articles without original data. A total of 14 primary studies were identified and included in the final analysis.

## 2.4 Data Extraction

Numerical data were extracted from each included study and organized into structured comparison tables. The following parameters were recorded for each study:

- Image guidance modality used
- Reported PTV margin size in millimeters
- Setup accuracy reported as mean displacement error in millimeters
- Intrafraction motion magnitude and correction threshold where reported
- Grade 2 or higher genitourinary and gastrointestinal toxicity rates
- Biochemical PSA control rates where reported

Data were extracted directly from tables, figures, and results sections of each paper. Where studies reported ranges, the midpoint value was used for comparative analysis.

## 2.5 Quantitative Analysis

### 2.5.1 Stochastic Motion Modeling

To quantify the time-dependent behavior of intrafraction prostate motion, a stochastic random walk model was implemented in Python using Jupyter Notebook. The model was based on the framework described by Ballhausen et al. (2015), in which prostate displacement is treated as a cumulative sum of small random movements over time.

The model was implemented as follows. At each time step, a small random displacement was drawn from a Gaussian distribution with zero mean and standard deviation, taken from parameters reported by Ballhausen et al. Displacement was accumulated over time to simulate realistic prostate drift during a treatment fraction. The simulation was run for treatment durations representative of both VMAT delivery (approximately 2 minutes) and SS-IMRT delivery (approximately 10 to 15 minutes). The probability of exceeding clinically relevant motion thresholds of 2mm, 3mm, and 5mm was calculated as a function of treatment duration. Each simulation was repeated across multiple iterations to produce statistically stable estimates. All modeling was performed in Python using the NumPy and Matplotlib libraries within Jupyter Notebook.

### 2.5.2 Statistical Outcome Analysis

To evaluate the relationship between PTV margin size and late gastrointestinal toxicity, a linear regression analysis was performed using data extracted from the included studies. Each study contributing both a margin value and a Grade 2+ GI toxicity rate was included as a data point. The Pearson correlation coefficient and corresponding p-value were calculated to assess the strength and significance of the relationship. This analysis was performed in Python using the SciPy library.

## 2.6 Limitations of the Method

Several limitations of this approach should be acknowledged. First, the included studies vary in sample size, follow-up duration, and reporting standards, which introduces heterogeneity into the comparison. Second, the stochastic model relies on parameters from a single source and may not fully capture the variability seen across different patient populations. Third, the regression analysis uses a small number of data points, which limits the statistical power of the correlation. These limitations are discussed further in Section 4.2.

# 3 Literature Review

## 3.1 Overview of Prostate Motion and Setup Uncertainties

Accurate dose delivery is important in prostate radiotherapy, and geometric uncertainties can affect it [2, 4]. Patient setup variability and internal organ motion create these uncertainties [2, 1]. Unlike rigid anatomical structures, the prostate continuously changes position because of physiological processes such as bladder filling, rectal gas, and patient relaxation [6]. Because of these variations, inter- and intra-fraction motion is introduced, posing fundamental challenges in delivering radiation precisely and reproducibly [2, 4, 6]. Interfraction motion refers to variations in prostate position between treatment fractions [2, 4]. These motions are primarily caused by daily changes in bladder and rectal volumes, as well as minor shifts in patient positioning [6]. Clinical studies report that inter-fraction prostate displacements generally range from 2 to 5 mm, but can occasionally exceed 10 mm, particularly in the anterior–posterior and superior–inferior directions [1]. Consequently, variations in prostate position can affect planning computed tomography scans and compromise frequent image guidance for accurate target localization [1]. Ballhausen et al. further modeled prostate motion using a random stochastic walk, treating prostate displacement as a sum of small, random movements over time [6]. This approach not only illustrates the observed increase in motion but also provides a quantitative framework for predicting intra-fraction motion [4]. Additionally, another study found that anatomical predictors, such as pelvic dimensions, correlate with increased prostate mobility, suggesting that individual-specific factors influence both movement trajectories and related uncertainties [2, 4]. Ballhausen et al. modeled prostate motion using a random-walk model. In which prostate displacement is treated as a cumulative sum of small random movements over time [6]. This model not only shows an observed increase in motion but also provides a quantitative framework for predicting intra-fraction motion [4, 2]. In another study, it was shown that anatomical predictors such as pelvic dimensions were correlated with increased prostate mobility. This suggested that individual-specific factors may influence movement trajectories and related uncertainties [2]. Clinical studies show that prostate motion creates a fundamental limitation to treatment precision [4, 2]. Both inter- and intra-fraction displacements compromise target dose coverage and can increase irradiation of surrounding organs at risk [1, 4]. This is especially significant in SBRT, where small margins and steep dose gradients are used. These effects can increase treatment-related toxicities and reduce tumor control [1, 2]. This underscores the need for accurate motion monitoring and management strategies [6, 2]. Understanding the size, timing, and physical characteristics of prostate motion is essential [4, 6]. This knowledge helps develop and optimize image-guided radiotherapy workflows [1, 2].

Research shows that unmanaged intrafraction motion leads to significant geometric

uncertainty. This highlights why continuous monitoring techniques are important.

## 3.2 Image Guidance Techniques for Prostate Radiotherapy

Image-guided radiotherapy (IGRT) plays a key role in prostate radiotherapy by reducing setup uncertainties and improving the accuracy of target localization [2, 4]. This allows for smaller margins and safer dose delivery [4, 5]. There are several IGRT techniques used in clinical practice, each differing in imaging method, accuracy, invasiveness, and workflow impact [2, 4].

### 3.2.1 Cone-Beam Computed Tomography (CBCT)

CBCT is an X-ray-based volumetric imaging technique integrated into many modern linear accelerators. It provides three-dimensional imaging immediately before treatment, enabling alignment of the prostate and surrounding anatomy with respect to the planning computed tomography scan [2, 4]. A major strength of CBCT is its ability to visualize anatomy in 3D and support daily position correction [2, 4]. However, repeated CBCT imaging contributes additional radiation dose, and CBCT is most commonly performed pre-treatment, which means intrafraction motion occurring during beam delivery may not be detected [2, 1]. In terms of workflow, CBCT increases treatment time due to image acquisition, registration, and couch correction steps [2, 4].

### 3.2.2 Fiducial Marker-Based Localization

Fiducial marker guidance involves implanting radiopaque markers, often gold seeds, in the prostate [2, 4]. These markers are then visualized with planar X-ray imaging to localize the prostate before treatment, and sometimes during delivery systems [1, 4]. Fiducial markers provide high localization accuracy and are widely used in prostate SBRT [4]. Nevertheless, implantation is invasive and may cause discomfort or procedural complications [2, 4]. In addition, fiducial tracking often relies on intermittent imaging, which limits continuous monitoring of intra-fraction motion and may still entail additional imaging dose [6, 4]. Workflow complexity is increased by the need for implantation, marker verification, and image-based matching procedures [2, 4].

These methods offer reliable positioning, but they cannot track continuous intrafraction motion. This limits their effectiveness for high-precision SBRT.

### 3.2.3 Magnetic Resonance Image-Guided Radiotherapy (MRIgRT)

MRI-guided radiotherapy combines radiation delivery with magnetic resonance imaging, offering excellent soft-tissue contrast and the potential for real-time visualization of the prostate and organs at risk [2, 4]. MRIgRT provides strong advantages for adaptive radiotherapy and motion management [2, 1]. However, it requires specialized equipment,

complex infrastructure, and substantial resources, which limits widespread availability and increases clinical workflow demands [2, 4].

#### 3.2.4 Electromagnetic (EM) Tracking

Electromagnetic tracking systems use implanted transponders that transmit position information to an external detector, enabling real-time prostate localization without ionizing radiation [2, 4]. The main advantage of EM tracking is continuous intrafraction monitoring with high temporal resolution [2, 6]. However, it requires implanted devices, which makes it invasive, and the system can increase costs and workflow complexity compared to standard imaging approaches [2, 4].

#### 3.2.5 Ultrasound Guidance

Ultrasound-guided radiotherapy uses non-ionizing imaging to directly visualize the prostate, helping with patient setup and sometimes motion monitoring [2, 4]. One main benefit is that ultrasound does not add extra imaging dose, so repeated imaging is possible [2, 1]. However, the quality of ultrasound images can depend on the operator and may change with patient anatomy or how the probe is positioned, which can affect how reproducible the results are [6, 4]. In terms of workflow, ultrasound can reduce the need for X-ray imaging, but staff need training and standard procedures to keep performance consistent [2, 4].

Real-time imaging technologies address the main limitation of static imaging by enabling continuous motion tracking. This helps make radiotherapy more adaptive and precise.

### 3.3 Ultrasound-Based Guidance: Principles and Technologies

Ultrasound-guided radiotherapy (USgRT) is becoming more popular as a non-invasive, radiation-free way to locate the prostate [2, 4]. Ultrasound uses high-frequency sound waves that reflect off tissue boundaries, letting clinicians see soft tissue structures in real time [4]. This makes it especially useful for prostate radiotherapy [2, 1]. Unlike X-ray imaging, ultrasound does not add extra ionizing radiation and can be used repeatedly or even continuously during treatment [2, 4].

Clinicians can use either transabdominal ultrasound (TAUS) or transperineal ultrasound (TPUS) for guidance. With TAUS, the probe is placed on the lower abdomen to image through the abdominal wall and bladder. This method is non-invasive and simple to use, but it is very sensitive to changes in bladder filling and how much pressure is applied with the probe. These factors can make images less consistent and add uncertainty [7, 8].

With TPUS, the probe is placed directly against the perineum, giving a view of the prostate from below. This setup keeps the probe more stable and is less affected by bladder filling. As a result, the images are more consistent and easier to reproduce [8, 9]. Because of these advantages, TPUS is now the preferred method for prostate radiotherapy, especially when high precision is needed, such as in SBRT.

Today's ultrasound guidance systems include three-dimensional imaging and real-time tracking. These tools let clinicians match ultrasound images directly with the planning CT scans. This helps them find any movement of the prostate and adjust the patient's position before treatment [7]. Continuous or almost continuous ultrasound can also track the prostate during treatment, so any significant movement can be spotted and corrected quickly if it goes beyond set limits [8].

Even with these benefits, ultrasound guidance has some technical challenges. Image quality can be affected by the patient's anatomy, shadows from tissues, and how the operator uses the equipment. These issues can impact how consistent and accurate the tracking is. Careful calibration and regular quality checks are needed to keep the system working well and to make sure images line up correctly [8]. Still, new technology and better clinical protocols are making ultrasound-guided prostate radiotherapy more reliable and useful in practice.

## 3.4 Setup Accuracy and Target Localization Using Ultrasound

Accurate patient setup and reliable target localization are essential for prostate radiotherapy. This is particularly important in stereotactic body radiotherapy, which uses small planning margins and steep dose gradients. Several studies have examined how ultrasound-based guidance systems help reduce setup uncertainties and improve geometric accuracy [2].

Arcadipane et al. used three-dimensional transabdominal ultrasound imaging to compare ultrasound-based localization with planning computed tomography. They found mean setup differences of 1.2 ± 2.3 mm, 1.5 ± 2.1 mm, and 1.8 ± 2.6 mm in the left-right, superior-inferior, and anterior-posterior directions, respectively [7]. These results showed good spatial agreement between ultrasound and computed tomography and confirmed that ultrasound-based positioning is feasible with clinically acceptable precision.

Richardson et al. studied the use of transperineal ultrasound guidance for daily prostate setup and found systematic setup errors below 2 mm in all three directions, with random errors usually between 1 and 3 mm [8]. They also observed that prostate displacements over 3 mm happened in more than half of treatment sessions, highlighting the need for daily image guidance to keep target localization accurate. Their results show that TPUS offers localization accuracy similar to fiducial markers and CBCT-based guidance, but without extra ionizing radiation exposure.

Higuchi et al. systematically studied how ultrasound-guided setup accuracy affects

treatment margin design. Using continuous ultrasound-based motion monitoring, they determined that planning target volume margins of 2-3 mm are enough to ensure proper target coverage and protect organs at risk when strong ultrasound guidance is used [5]. These findings are especially important for SBRT, where reducing margins helps protect healthy tissues.

Overall, these studies demonstrate that ultrasound-guided prostate localization consistently achieves setup accuracy of 1–3 mm. This level of precision supports the clinical feasibility of small-margin treatment planning and provides a strong foundation for the safe implementation of high-dose hypofractionated prostate radiotherapy [6].

**Table 1:** Summary of reported setup accuracy and margin recommendations for ultrasound-guided prostate radiotherapy.

| Study | Technique | Setup Accuracy (mm) | Recommended Margin (mm) |
|---|---|---|---|
| Arcadipane et al. (2015) | 3D TAUS | 1–3 mm | 3–5 mm |
| Richardson et al. (2017) | TPUS | 1–3 mm | 2–3 mm |
| Higuchi et al. (2022) | TPUS | 1–2 mm | 2–3 mm |

## 3.5 Intrafraction Motion Monitoring and Validation

Intrafraction prostate motion is a major source of geometric uncertainty in prostate radiotherapy, especially in stereotactic body radiotherapy, which uses high doses per fraction and steep dose gradients. Real-time monitoring of the prostate during treatment is essential to keep the target covered and reduce radiation to nearby organs. Researchers have studied ultrasound-based tracking systems for this purpose, as they give quantitative information about prostate motion and support dynamic motion management.

Researchers have used transperineal ultrasound to measure how much and how often the prostate moves during treatment. Richardson et al. found that prostate shifts of more than 3 mm happened in about 52% of treatment sessions, with some shifts going over 6 mm [8]. Guo et al. also found that prostate movement greater than 2 mm occurred during about 7 percent of the time when the radiation beam was on, showing that these movements, though not constant, are still important [9]. These results show that even short SBRT treatments can have prostate motion that goes beyond standard planning margins.

**Table 2:** Representative studies on intra-fraction prostate motion and monitoring strategies, summarizing methods, cohorts, key findings, and limitations.

| Study | Monitoring method | Cohort / dataset | Key findings | Limitations |
|---|---|---|---|---|
| Ballhausen et al. (2015) | 4D Transperineal Ultrasound (Elekta Clarity) | 6 patients, 84 fractions | Intrafraction motion followed a random-walk behaviour with positional variance increasing over time; fixed margins may be suboptimal. | Substantial inter-patient variability; large excursions observed in some fractions. |
| Guo et al. (2023) | Transperineal Ultrasound (Clarity Autoscan), SBRT, VMAT | 23 patients, 110 fractions | Prostate displacement exceeded 2 mm in 21% of fractions; ultrasound enabled real-time motion detection during SBRT delivery. | Motion patterns were unpredictable; rectal preparation was inconsistent across patients. |
| Higuchi et al. (2022) | Ultrasound image-guided radiotherapy (TPUS), SS-IMRT vs. VMAT | 14 patients, 490 fractions | Shorter VMAT delivery times reduced intrafraction motion compared with SS-IMRT, supporting time-dependent margin considerations. | Small cohort size; single-institution study. |
| Baker et al. (2016) | 4D Transperineal Ultrasound with fiducial comparison | 10 patients, 51 fractions | Prostate motion was generally small during short treatments, with posterior motion predominance commonly observed. | Limited sample size; rotational motion was not assessed. |

As summarized in Table 2, intrafraction prostate motion is time-dependent and patient-specific across monitoring techniques.

Ballhausen et al. studied how prostate motion changes over time and found that displacement increases as treatment goes on [6]. They reported that in the first few minutes, the prostate usually moves 1 to 2 mm, but with longer treatments, the movement often goes over 3 mm. This increase is linked to factors like muscle relaxation, rectal movement, and changes in organ filling, which shows why it is important to keep treatments short and monitor motion continuously.

Ballhausen et al. used a random walk model to describe how prostate motion changes randomly over time [6]. This model matched the observed increase in motion variation as treatment lasted longer and helped predict how much the prostate might move. Later, they found that factors like pelvic size were linked to greater motion, suggesting that each patient's anatomy affects how much the prostate moves [10]. These results point to the possibility of tailoring motion management to each patient.

Besides studying how the prostate moves, researchers have tested correction strategies that use real-time ultrasound monitoring. In these methods, if the prostate moves more than a set amount (usually 2 to 3 mm), treatment is paused and the patient is repositioned. Guo et al. showed that using a 2 mm threshold greatly reduced leftover motion during treatment, improving accuracy without causing too many interruptions [9]. These findings show that real-time motion correction is practical and that ultrasound-guided monitoring is valuable for SBRT.

In summary, research shows that intrafraction prostate motion depends on both time and the individual patient, and significant movement often happens during treatment. Ultrasound-based monitoring is an effective way to detect and manage this motion in real time. It allows for correction strategies that improve treatment accuracy and make it possible to safely reduce margins in prostate SBRT.

The reported intrafraction motion magnitudes and correction thresholds are summarized in Table 3.

**Table 3:** Summary of reported intrafraction prostate motion and correction strategies using ultrasound guidance.

| Study | Technique | Motion Magnitude | Correction Threshold |
|---|---|---|---|
| Richardson et al. (2017) | TPUS | $> 3$ mm in 52% of fractions | 3 mm |
| Guo et al. (2023) | TPUS | $> 2$ mm in 7% of beam-on time | 2 mm |
| Ballhausen et al. (2015) | TPUS | 1–3 mm typical drift | Not applied |
| Ballhausen et al. (2023) | TPUS | Anatomy-dependent motion | Not applied |

## 3.6 Clinical Outcome Evidence of Ultrasound-Guided Prostate Radiotherapy

To properly evaluate ultrasound-guided radiotherapy, it is important to consider patient outcomes such as tumor control, PSA response, and treatment-related side effects. Clinical studies have found that this method produces positive results that are similar to other well-established image-guided techniques [7].

Takai et al. reported on patients treated with transperineal ultrasound-guided radiotherapy and found strong biochemical control and stable PSA levels during long-term follow-up [1]. Most patients experienced lasting decreases in PSA, which indicates good tumor control. These findings support that ultrasound is reliable for accurate targeting and dose delivery [1].

Ultrasound-guided radiotherapy has also shown positive results regarding side effects for both genitourinary and gastrointestinal health. Takai et al. observed few moderate or severe side effects, and most patients only had mild or short-term urinary and rectal symptoms [1]. These rates are similar to those with CBCT-guided and fiducial-based treatments, suggesting that ultrasound guidance is safe for patients.

Researchers have also studied ultrasound-guided stereotactic body radiotherapy (SBRT). Lee et al. found that SBRT achieved high local control and acceptable side effects, which supports using shorter treatment courses for prostate cancer [4]. With precise image guidance, SBRT can reduce treatment times while still providing strong tumor control and manageable side effects. These results support adding ultrasound-based guidance to prostate SBRT workflows and show its potential to deliver accurate and patient-friendly radiotherapy [11].

**Table 4:** Summary of reported clinical outcomes for ultrasound-guided prostate radiotherapy.

| **Study** | **Treatment** | **Tumor Control / PSA Response** | **GU/GI Toxicity** |
|---|---|---|---|
| Takai et al. (2024) | TPUS-guided RT | High biochemical control, stable PSA | Low grade $\geq$2 toxicity |
| Lee et al. (2021) | SBRT | High local control | Acceptable toxicity |

## 3.7 Summary and Identified Knowledge Gaps

The research discussed in this chapter shows that ultrasound-guided radiotherapy has been thoroughly tested for treating prostate cancer. Many studies confirm it can accurately position patients, locate targets, and monitor movement during treatment. Data also show that ultrasound guidance can achieve accuracy within a few millimeters, which allows for smaller treatment margins and better protection of nearby organs. Overall, these results support that ultrasound-based image guidance is technically sound and reliable, especially for precise treatments like stereotactic body radiotherapy.

Clinical studies also show that ultrasound-guided prostate radiotherapy is both safe and effective. Tumor control rates, PSA responses, and side effect profiles are similar to those seen with other established image-guided radiotherapy methods. This suggests that ultrasound-based guidance can be used in clinical practice without reducing treatment success or patient safety.

Even with strong technical and clinical evidence, some important questions remain. Most studies focus on accuracy and motion tracking, but fewer look at how these improvements affect patient outcomes. There are also not many direct comparisons between ultrasound-guided SBRT and other image-guided radiotherapy methods, especially studies that evaluate accuracy, motion control, and patient results together.

These limitations highlight the need for comprehensive clinical studies that integrate technical performance metrics with long-term patient outcome data across multiple institutions. These gaps show that more thorough clinical studies are needed, combining technical performance with patient outcomes. This work aims to synthesize evidence from published studies on ultrasound-guided SBRT for prostate cancer, evaluating treatment accuracy, organ motion control, and clinical outcomes through a structured systematic review and quantitative analysis. By comparing ultrasound-guided methods with other image-guided techniques, this study hopes to better define the clinical role and benefits of ultrasound guidance in prostate SBRT, helping to fill current gaps and improve treatment practices.

# 4 Quantitative and Clinical Comparison of Radiotherapy Techniques

In this chapter, we will discuss a comparison of different techniques used in prostate radiotherapy. The main goal is to determine how various approaches affect treatment accuracy and patient outcomes.

Various factors can influence the quality of treatment. For example: the treatment machine, margin calculation methods, the type of imaging used, and the use of adaptive radiotherapy. Each of these plays an important role in reducing uncertainties during treatment.

Many researchers have reported extensive results. Some techniques focus on reducing dose delivery time or improving workflow, whereas some are better at accuracy. To determine which technique is best, we need to compare techniques and methodologies using both numerical data and clinical outcomes.

Overall, in this chapter, we will discuss a structured comparison based on the available data. Comparing parameters such as dose delivery amount, margin size, and organ motion will be analyzed. The aim is to find out which techniques will provide safer treatment and greater precision.

After this comparison, we will have a clear set of highlights on the strengths and limitations of each method. Not only this, but it will also provide a basis for more effective strategies in modern radiotherapy practices.

In this study, reported clinical data from the literature are not only reviewed but also systematically analyzed to evaluate the practical impact of different motion management strategies in prostate SBRT.

## 4.1 Comparison of PTV Margin Strategies

The evolution of PTV margin strategies in prostate SBRT reflects a fundamental change in how geometric uncertainty is conceptualised in radiotherapy. Rather than being treated as a fixed geometric safety buffer, the PTV margin increasingly represents the residual uncertainty that remains after all available motion management strategies have been applied.

In early radiotherapy workflows, large population-based margins in the range of 7–10 mm were required to ensure tumour coverage under conditions of limited imaging and unmonitored intrafraction motion [7, 11, 12]. These margins were derived from worst-case assumptions rather than patient-specific motion data, meaning that target coverage was guaranteed at the cost of intentionally irradiating larger volumes of healthy tissue. This reflects a planning philosophy in which robustness was prioritised over anatomical precision, as motion could not be reliably observed or corrected during delivery.

The introduction of image-guided radiotherapy marked a first refinement of this approach. Techniques such as CBCT-based fiducial tracking reduced setup uncertainty and enabled moderate margin reduction to approximately 7 mm [13]. However, this improvement primarily addressed inter-fraction positioning errors rather than intrafraction motion. Consequently, prostate displacement occurring during beam delivery remained largely uncorrected, meaning that the fundamental limitation of time-dependent motion was not resolved by imaging alone.

A more substantial shift occurs with the adoption of real-time imaging and adaptive delivery systems, including Transperineal Ultrasound (TPUS) and gated SBRT workflows. In these approaches, margins are reduced to approximately 2–5 mm because intrafraction motion is no longer assumed but actively monitored and managed during treatment delivery [4, 1]. This represents a conceptual transition from compensating for unknown motion to directly measuring and controlling motion in real time.

Ultra-small margins in the range of 2–3 mm further extend this paradigm by treating residual motion as a bounded stochastic process rather than an unobserved uncertainty. In these systems, beam delivery can be paused or corrected when predefined thresholds are exceeded, meaning that margins function as a safety tolerance rather than the primary mechanism for motion compensation [9]. This decouples margin size from worst-case geometric assumptions and redefines it as a parameter dependent on monitoring fidelity.

The data synthesised in Table 5 supports a consistent inverse relationship between imaging sophistication and required PTV margin size. However, this trend should be interpreted as a transition in uncertainty management strategy rather than a simple technological progression. The key driver is not only improved image quality, but the ability to actively observe and respond to intrafraction motion during irradiation.

Clinically, this transition has direct consequences for normal tissue sparing. Reduced margins decrease the volume of rectum and bladder included in the high-dose region, thereby lowering the probability of radiation-induced toxicity while maintaining tumour coverage. This indicates that margin reduction is not merely a planning optimization but a direct surrogate marker of the level of real-time motion control achieved in a given treatment workflow.

Overall, the progression from 10 mm to 2–3 mm margins represents a shift from static, uncertainty-driven planning toward motion-managed radiotherapy, where treatment precision is increasingly governed by continuous anatomical feedback rather than fixed geometric assumptions.

#### 4.1.1 The Evolution from Static to Dynamic Margin Paradigms

The evolution of PTV margin design reflects a fundamental redefinition of geometric uncertainty in prostate SBRT. Rather than functioning as a fixed geometric safety expansion, the PTV margin increasingly represents the residual uncertainty that remains

after all available motion management strategies have been applied during treatment delivery.

This shift can be understood as a transition from uncertainty compensation to uncertainty reduction. In early radiotherapy workflows, intrafraction motion was effectively unobservable, which necessitated large population-based margins in the range of 7–10 mm to ensure target coverage under worst-case assumptions [7, 11, 12]. These margins were not patient-specific but statistically conservative, meaning that geometric safety was achieved by deliberately expanding irradiation volumes. The consequence of this strategy was an unavoidable increase in dose exposure to surrounding organs at risk, particularly the rectum and bladder, indicating that early treatment design prioritised robustness over anatomical precision.

The introduction of image-guided radiotherapy introduced a first-order refinement in this framework by improving pre-treatment localisation. Fiducial-based CBCT alignment reduced setup uncertainty and enabled intermediate margin reductions to approximately 7 mm [13]. However, this improvement remained fundamentally limited because it operated on a pre-treatment imaging paradigm. In this model, anatomy is assumed to remain stable during irradiation, meaning that intrafraction motion is not directly measured or corrected. As a result, residual uncertainty persists as an unobserved variable during beam delivery.

The comparative performance of different guidance strategies is summarised in Table 5. The data demonstrates a consistent stepwise reduction in PTV margins as imaging and tracking capability improves:

**Table 5:** Comparative Analysis of PTV Margin Strategies and Associated Modalities

| **Source** | **Guidance System** | **Margin (mm)** | **Clinical Rationale** |
|---|---|---|---|
| Richardson et al. (2017) | Standard IGRT | 10 mm | High-confidence coverage under non-monitored delivery conditions |
| Arcadipane et al. (2015) | 3D-US / CT | 8–10 mm | Large margins required due to absence of real-time correction |
| Fonteyne et al. (2011) | Fiducial + CBCT | 7 mm | Improved setup accuracy but no intrafraction correction |
| Lee et al. (2021) | CBCT + TPUS | 5 mm | Hybrid imaging reduces uncertainty through partial real-time feedback |
| Takai et al. (2024) | Real-time TPUS | 3–5 mm | Continuous monitoring enables reduction of posterior margin |
| Guo et al. (2023) | Gated SBRT | 2–3 mm | Motion-triggered beam interruption replaces geometric expansion |

The trend observed in Table 5 is not merely a reflection of improved imaging quality, but indicates a structural change in how uncertainty is managed. Each step reduction in margin size corresponds to the introduction of a new level of temporal information about prostate motion. In non-adaptive systems, uncertainty is handled through geometric expansion, whereas in modern systems it is progressively replaced by direct observation and control of motion during irradiation.

This interpretation is reinforced by the transition from CBCT-based workflows to real-time ultrasound and gating-based systems. In these later modalities, margins are no longer designed to compensate for unknown motion but instead represent the residual motion that persists despite active monitoring. This fundamentally alters the function of the PTV margin from a primary safety mechanism to a secondary tolerance parameter within an actively controlled system.

The implications of this shift are illustrated in Figure 2, which shows a clear progression from large-margin, non-adaptive techniques toward small-margin, real-time guided systems. The figure highlights that the most significant reductions occur only when continuous motion feedback is introduced, confirming that temporal resolution of motion, rather than spatial imaging quality alone, is the key determinant of margin reduction.

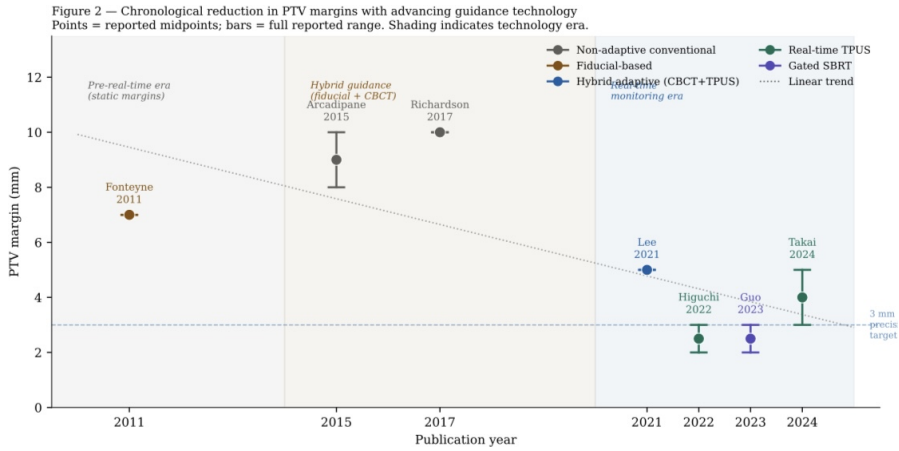


**Figure 2:** Comparison of PTV margins across different imaging techniques.

Clinically, this reduction in margin size has direct consequences for dose distribution. Smaller margins reduce irradiation of adjacent organs at risk, particularly the rectum and bladder, while maintaining target coverage due to improved motion control. This demonstrates that margin reduction is not simply a planning optimisation, but a surrogate indicator of how effectively intrafraction motion is managed within a given radiotherapy workflow.

Overall, the progression from 10 mm to 2–3 mm margins reflects a fundamental transition in radiotherapy philosophy: from static, worst-case geometric safety design to dynamic, feedback-driven motion management. In this modern framework, treatment precision is no longer primarily determined by pre-treatment assumptions, but by the system's ability to continuously observe and control anatomy during dose delivery.

### 4.1.2 Mathematical Modeling of Intrafraction Motion: The Random Walk

A more mechanistic understanding of intrafraction prostate motion requires moving beyond static geometric uncertainty toward stochastic temporal models. In this context, intrafraction displacement has been widely modeled as a stochastic diffusion process, often approximated as a Random Walk or Brownian motion [6, 5]. This framework provides a time-dependent description of motion that directly contrasts with conventional static error assumptions used in traditional margin recipes.

A key implication of this model is that motion uncertainty is not constant during beam delivery but increases with treatment duration. Specifically, the intrafraction variance can be expressed as a function of time:

- **Time-dependent uncertainty growth:** Intrafraction motion is characterized by a progressive increase in positional variance over time, such that $\sigma_{it} \propto \sqrt{t}$. This implies that even small-amplitude stochastic motion becomes clinically relevant when delivery times are prolonged.

- **Clinical implication for delivery techniques:** This temporal scaling provides a physical justification for the superiority of fast delivery techniques such as VMAT over conventional step-and-shoot IMRT. By reducing beam-on time (typically $\sim$2 minutes for VMAT versus 10–15 minutes for IMRT), the cumulative probability of exceeding clinically relevant displacement thresholds (e.g., 2–3 mm) is significantly reduced.

- **Connection to margin formalism:** Classical margin recipes, such as the van Herk formulation, assume separable systematic and random components:

#### 4.1.3 Stochastic Modeling of Intrafraction Prostate Motion

To quantitatively evaluate intrafraction prostate motion, a stochastic random walk model was implemented in Python using a Jupyter Notebook, based on motion characteristics reported by Ballhausen et al. [6]. Prostate displacement was modeled as a cumulative sum of normally distributed positional increments, where each increment represents the net displacement occurring within a one-second interval of beam delivery. The standard deviation of each increment was set to $\sigma = 0.18$ mm/s, consistent with step-wise motion estimates reported in the literature [6].

A Monte Carlo simulation framework was used to generate 5000 independent motion trajectories for treatment durations ranging from 60 to 300 seconds, sampled at 10-second intervals. For each trajectory, the maximum absolute displacement from the initial position was computed. The probability of exceeding clinically relevant displacement thresholds (2 mm, 3 mm, and 5 mm) was estimated as the proportion of trajectories surpassing each threshold.

To ensure numerical stability, convergence was assessed by comparing simulation runs with 1000, 2000, and 5000 trajectories at a fixed duration of 120 seconds and a 2 mm threshold, yielding probabilities of 0.583, 0.585, and 0.562 respectively. The small variation confirms adequate convergence at 5000 iterations. A fixed random seed was used to ensure full reproducibility of all simulations.

The results are presented in Figure 3. The probability of exceeding a 2 mm displacement increases from approximately 0.05 at 60 seconds to approximately 0.90 at 300 seconds. For the 3 mm threshold, the probability increases from approximately 0.05 to 0.60 over the same interval, while the 5 mm threshold shows a slower rise, reaching approximately 0.20 at 300 seconds.

At clinically relevant delivery times, marked by the vertical reference lines, the probability of exceeding a 2 mm displacement is approximately 0.56 at 120 seconds (typical VMAT delivery) and exceeds 0.90 at 300 seconds. This highlights the strong dependence of intrafraction motion on treatment duration and emphasizes the increased geometric risk associated with prolonged delivery techniques such as step-and-shoot IMRT.

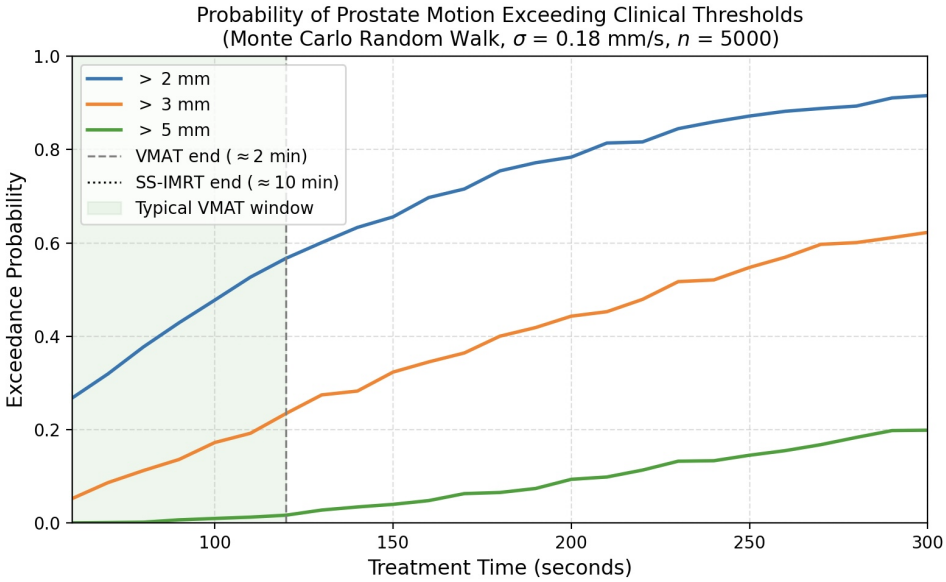


**Figure 3:** Probability of prostate displacement exceeding clinical thresholds of 2 mm, 3 mm, and 5 mm as a function of treatment duration, estimated via Monte Carlo random walk simulation ($n = 5000$, $\sigma = 0.18$ mm/s, based on Ballhausen et al. 2015). Vertical dashed lines indicate typical VMAT (≈120 s) and SS-IMRT (≈600 s) delivery durations. The shaded region highlights the typical VMAT delivery window.

These results indicate that intrafraction prostate motion is inherently stochastic and strongly time-dependent. The progressive increase in exceedance probability demonstrates that geometric uncertainty accumulates during beam delivery, making treatment duration a critical factor in motion management. Shorter delivery techniques such as VMAT reduce—but do not eliminate—this uncertainty by limiting the temporal window for motion accumulation.

Overall, this analysis demonstrates that intrafraction motion cannot be reliably managed using fixed safety margins alone. Instead, the time-dependent probability structure provides a quantitative justification for rapid delivery techniques and reinforces the clinical importance of real-time motion monitoring in prostate SBRT.

#### 4.1.4 Duration-Dependent Margins and the "Random Walk"

Improving the accuracy of SBRT margin design requires moving beyond static uncertainty models toward formulations that explicitly incorporate treatment duration. As demonstrated in recent studies, prostate positional variance increases with time in a manner consistent with stochastic diffusion processes, commonly described as a Random Walk or Brownian motion model [1, 5]. This implies that geometric uncertainty is not fixed, but accumulates progressively during beam delivery, necessitating duration-dependent

margin strategies.

- **Technique dependence of margin requirements:** For rapid VMAT delivery arcs (typically $< 3$ minutes), a margin on the order of 2 mm may provide sufficient geometric confidence. In contrast, for prolonged step-and-shoot IMRT sessions ($> 10$ minutes), larger margins approaching 5 mm are required to maintain target coverage reliability under increasing motion uncertainty [5, 6].
- **Time-dependent stability degradation:** Clinical evidence indicates that approximately 88% of treatment fractions exhibit vector displacements below 2 mm during short treatment sessions; however, this stability deteriorates as treatment duration increases, reflecting cumulative stochastic drift of prostate position over time [1].
- **Margin formalism refinement:** Recent work suggests that Restricted Maximum Likelihood (REML)-based approaches provide improved estimation of motion uncertainty compared to traditional van Herk formulations, particularly in SBRT settings. By reducing systematic overestimation of geometric variance, REML enables more accurate derivation of tight margins appropriate for high-gradient dose distributions [1, 5].

Historically, PTV margins in the range of 7–10 mm were routinely employed as conservative safety buffers to account for unobserved intrafraction motion. In modern SBRT practice, however, such large expansions are no longer compatible with the steep dose gradients required for prostate treatments. Instead, reduced margins in the range of 2–5 mm are achieved through a combination of real-time imaging, motion tracking, and adaptive delivery strategies. This reduction is not merely a technical refinement but a clinical necessity, as it directly contributes to lowering radiation exposure to adjacent organs at risk, particularly the rectum and bladder, thereby reducing the incidence of Grade $\geq$2 toxicities [1, 4].

These findings demonstrate that static margin strategies are insufficient to capture the time-dependent nature of intrafraction motion. Effective margin design must therefore incorporate duration-aware principles that explicitly account for treatment delivery time, particularly in prostate SBRT where high dose gradients amplify the clinical impact of even sub-millimetric geometric uncertainties.

## 4.2 Imaging Techniques and Setup Accuracy

The accuracy of target localization in radiotherapy is fundamentally dependent on the quality of imaging used for patient positioning and verification. In modern prostate SBRT workflows, advanced Image-Guided Radiotherapy (IGRT) plays a central role in reducing geometric uncertainty by improving the visualization of both the target volume

and surrounding organs at risk (OARs). This enhanced anatomical definition enables more precise correction of both systematic and random setup errors, thereby directly supporting margin reduction strategies [1, 7].

Unlike conventional radiotherapy, where setup corrections were primarily based on external anatomical landmarks or periodic imaging, modern IGRT systems provide direct or near-real-time assessment of internal soft-tissue position. This shift is particularly important in prostate treatments, where inter-fraction variability and organ deformation can significantly affect target localization. As a result, setup accuracy is no longer treated as a static pre-treatment adjustment but as a continuously managed geometric parameter throughout the treatment course.

The improvement in imaging capability has therefore redefined the relationship between geometric uncertainty and PTV margin design. Higher-quality imaging reduces reliance on population-based safety expansions and enables more individualized, anatomy-driven correction strategies.

### 4.2.1 Correlation Between Modality and Localization Precision

The transition from static, image-based localization to continuous or semi-continuous imaging represents a fundamental improvement in prostate SBRT setup accuracy. Traditional Cone-Beam Computed Tomography (CBCT) provides a volumetric "snapshot" of patient anatomy prior to treatment delivery. However, because CBCT is typically registered using either bony anatomy or implanted fiducial markers, it does not fully capture soft-tissue deformation or intrafraction motion, which can introduce residual uncertainties requiring larger safety margins [3].

In contrast, Transperineal Ultrasound (TPUS) enables direct visualization of the prostate and surrounding soft tissues in real time. This improves localization of the prostate–bladder interface and reduces dependency on surrogate anatomical structures. As a result, TPUS-based workflows support tighter margins in the range of 2–5 mm, particularly in SBRT settings where high dose gradients make geometric precision critical [7, 12].

The quantitative comparison of imaging modalities is summarized in Table 6, which consolidates setup accuracy metrics reported across multiple clinical studies.

**Interpretation of Table 6:** The data demonstrate a consistent trend in which imaging systems with improved soft-tissue visualization and real-time tracking capability achieve higher localization precision (approximately 1–2 mm). CBCT-based approaches show slightly larger residual uncertainties due to reliance on surrogate alignment strategies, whereas TPUS and hybrid CBCT+TPUS systems consistently achieve sub-2 mm accuracy. Importantly, this reduction in setup error directly translates into the clinical feasibility of smaller PTV margins in SBRT planning.

From a clinical perspective, this trend reflects a transition from passive geometric correction (pre-treatment alignment only) toward active motion-aware localization

strategies that continuously or repeatedly verify target position. This shift has direct implications for both treatment accuracy and normal tissue sparing, particularly in rectal and bladder dose reduction.

**Table 6:** Comparison of Imaging Modalities and Setup Accuracy Metrics

| **Study** | **Imaging Type** | **Accuracy** | **Clinical Observation** |
|---|---|---|---|
| Li et al. (2015) | 3D TPUS vs. CBCT | 1.3–1.4 mm | TPUS accuracy is comparable to seed-based CBCT for prostate positioning [11]. |
| Arcadipane et al. (2015) | 3D US / CT | 1.2–2.6 mm | US/CT fusion improved target delineation in metal artifact cases [7]. |
| Richardson et al. (2017) | 4D TPUS | $<$ 2.0 mm | Systematic setup errors maintained below 2 mm in all directions [8]. |
| Lee et al. (2021) | CBCT + TPUS | 1.0–2.0 mm | Hybrid imaging provides high-precision tracking for SBRT [4]. |
| Han et al. (2018) | 4D TPUS | 1.1–3.3 mm | US tracking shows high agreement with fiducial positions [2]. |

### 4.2.2 The Impact of Operator Experience and the Learning Curve

While imaging modality plays a central role in localization accuracy, operator-dependent variability remains a critical secondary factor influencing overall setup precision in prostate SBRT. Unlike automated registration processes, ultrasound-based systems such as TPUS require manual interpretation of anatomical boundaries, particularly the prostate–bladder and prostate–rectum interfaces, which introduces inter-observer variability.

- **Inter-observer Variability:** Evidence shows that inexperienced operators may inconsistently identify soft-tissue boundaries, resulting in systematic setup deviations of several millimeters. These errors can partially negate the theoretical accuracy advantages of advanced imaging systems if not properly controlled.
- **The Learning Curve:** Grimwood et al. (2021) demonstrated that operator precision improves significantly after structured training and repeated clinical exposure. Standardized imaging protocols, combined with automated or

semi-automated contouring tools (e.g., Clarity systems), reduce variability and improve reproducibility across treatment fractions [6, 12].

- **Clinical Implication:** In high-dose SBRT workflows where margins are typically 2–3 mm, even small operator-dependent errors can become clinically significant. Therefore, early-phase implementation of TPUS-based workflows should incorporate double-verification protocols or expert review to ensure safe and consistent target localization.

The learning curve effect highlights that imaging accuracy is not solely a hardware-dependent parameter but a combined function of technology, operator skill, and institutional protocol maturity. Consequently, achieving sub-3 mm margin safety requires both advanced imaging systems and standardized training pathways to fully realize the benefits of motion-aware radiotherapy.

#### 4.2.3 Technical Considerations in Ultrasound-Based Guidance

Although ultrasound-based guidance is non-invasive and well-suited for real-time prostate tracking, its accuracy is still influenced by several physical and procedural factors that determine its clinical reliability in SBRT workflows.

- **Probe Pressure:** In transperineal ultrasound (TPUS), the probe is externally stabilized and applies minimal and relatively constant pressure. This results in negligible prostate displacement, typically below 1 mm, which is significantly lower than the deformation risk associated with transabdominal ultrasound approaches. This mechanical stability is a key factor enabling sub-millimetric reproducibility in TPUS-based localization [5, 12].

- **Real-time Monitoring:** Unlike Cone-Beam Computed Tomography (CBCT), which provides only pre- and post-treatment anatomical snapshots, TPUS enables continuous volumetric monitoring at approximately 2 Hz. This allows the system to capture transient intrafraction motions such as drift or micro-displacements that occur during beam delivery and would otherwise remain unobserved in conventional imaging workflows [13, 12].

Although both CBCT-based and ultrasound-based systems can achieve comparable setup accuracy at the time of patient positioning, their fundamental difference lies in temporal resolution. CBCT is inherently a static imaging modality, whereas TPUS introduces a dynamic feedback loop that continuously updates prostate position during treatment delivery. As a result, TPUS does not primarily improve initial alignment accuracy but rather enhances motion management capability throughout irradiation.

The results demonstrate that imaging modalities with similar geometric setup precision can yield substantially different clinical outcomes depending on their ability

to manage intrafraction motion. Conventional imaging techniques are limited by their snapshot-based nature, which prevents detection of time-dependent displacement. In contrast, real-time ultrasound monitoring enables active motion awareness, which is essential for maintaining target coverage in high-dose SBRT treatments with tight (2–3 mm) margins. This highlights that modern radiotherapy accuracy is increasingly governed not only by spatial resolution, but by temporal sampling capability and continuous motion surveillance.

## 4.3 Machine-Based Comparison and Technology Impact

The selection of a treatment delivery platform is a critical determinant of achievable precision in prostate radiotherapy, as each system inherently defines the balance between imaging frequency, motion management capability, and dose delivery speed. In modern Stereotactic Body Radiotherapy (SBRT), high geometric accuracy is no longer achieved solely through pre-treatment setup, but through the integration of rapid dose delivery with repeated or continuous target verification during irradiation. This combined capability is essential for mitigating dosimetric uncertainty arising from intrafraction prostate motion [1, 13, 9].

### 4.3.1 Technological Influence on Targeting and Efficiency

Radiotherapy platforms differ substantially in the degree of integration between imaging systems and beam delivery, and this integration directly determines how effectively intrafraction motion can be managed during treatment. Conventional C-arm linear accelerators typically rely on Cone-Beam Computed Tomography (CBCT) for pre-treatment positioning, providing a single volumetric verification step prior to irradiation. While this approach improves setup accuracy, it does not allow continuous adaptation during dose delivery [7].

In contrast, advanced systems such as MR-Linac platforms or 4D Transperineal Ultrasound (TPUS)-guided systems incorporate real-time or near-real-time imaging during beam delivery. This enables continuous or high-frequency monitoring of prostate position, allowing motion to be detected and managed while treatment is ongoing rather than only before irradiation begins [7, 13, 2].

In parallel with these imaging advancements, improvements in delivery techniques have significantly reduced treatment time. The transition from Step-and-Shoot IMRT to Volumetric Modulated Arc Therapy (VMAT) has reduced typical beam-on times from approximately 10–20 minutes to under 3 minutes. This reduction is clinically important because shorter treatment durations directly limit the temporal window available for intrafraction prostate motion, thereby reducing the probability of clinically significant displacement during irradiation [13, 12].

Overall, the combined evolution of faster delivery techniques and integrated real-time imaging systems represents a shift from static, pre-treatment verification toward dynamic, motion-aware radiotherapy, in which targeting accuracy is maintained throughout the entire treatment process rather than only at its initiation.

### 4.3.2 Comparative Analysis of Treatment Platforms

Table 7 synthesizes reported performance metrics across different radiotherapy delivery platforms, highlighting how increasing integration between imaging and beam delivery influences achievable targeting precision in prostate SBRT.

**Table 7:** Comparative Performance of Radiotherapy Delivery Platforms

| **Study** | **Machine / System** | **Precision (mm)** | **Clinical Observation** |
|---|---|---|---|
| Fonteyne et al. (2011) | Conventional Linac | 3–5 mm | Reliance on gold fiducial markers for setup; lacks real-time intrafraction tracking [13]. |
| Arcadipane et al. (2015) | Linac + 3D-US | 1.2–2.6 mm | Enhanced soft-tissue localization enables accurate targeting despite CT metal artifacts [7]. |
| Richardson et al. (2017) | Linac + 4D-TPUS | $< 2.0$ mm | First UK implementation of the Clarity system; confirmed sub-2 mm systematic setup errors [8]. |
| Lee et al. (2021) | Linac + TPUS/CBCT | 1.0–2.0 mm | High-precision hybrid guidance supports safe ultrahypofractionated SBRT delivery [4]. |
| Takai et al. (2024) | Linac + CAS (TPUS) | 1.0–3.0 mm | Real-time tracking enabled 3 mm posterior margins with maintained biochemical control [1]. |

The data demonstrate a clear trend in which increasing integration between imaging and treatment delivery systems is associated with progressively improved targeting precision. Conventional linac-based workflows relying on fiducial markers achieve moderate accuracy (3–5 mm) but remain limited by the absence of intrafraction motion monitoring. In contrast, systems incorporating ultrasound-based guidance or hybrid

imaging consistently achieve sub-3 mm precision by enabling improved soft-tissue visualization and more frequent positional verification.

Importantly, 4D-TPUS and hybrid TPUS/CBCT systems demonstrate that high geometric precision is not solely dependent on spatial imaging resolution, but also on the temporal capability to track motion during delivery. The most advanced systems (e.g., CAS-guided TPUS workflows) extend this principle further by enabling real-time correction, allowing for tighter posterior margins without compromising treatment safety.

Overall, the table supports a progressive transition from static, pre-treatment alignment paradigms toward dynamically monitored radiotherapy systems in which targeting accuracy is maintained throughout the entire delivery process rather than at a single time point.

### 4.3.3 Clinical Advantages of Fiducial-Free Real-Time Tracking

A central question in modern prostate SBRT is whether real-time intrafraction tracking should rely on invasive electromagnetic (EM) transponders or non-invasive systems such as Transperineal Ultrasound (TPUS). Although both approaches provide continuous or near-continuous positional information, fiducial-free workflows offer distinct clinical, biological, and logistical advantages that extend beyond purely geometric considerations.

- **Mitigation of Procedural Risks:** Invasive fiducial systems, including gold seed markers and Calypso electromagnetic transponders, require implantation via transrectal or transperineal procedures. Evidence from Biston et al. (2019) and Fonteyne et al. (2011) reports an approximate 20% incidence of minor procedure-related complications, such as hematuria, hematospermia, and transient prostatitis [14, 13]. Although generally self-limiting, these risks are entirely avoided in non-invasive TPUS-based workflows.

- **Elimination of Marker Migration:** A well-documented limitation of fiducial-based image guidance is marker migration, where implanted seeds shift relative to the surrounding prostate tissue over time. This introduces a surrogate-based error that may not accurately reflect true target motion. TPUS circumvents this limitation by directly visualizing the soft-tissue interface of the prostate, ensuring that the tracking signal is anatomically representative of the target itself [7].

- **Workflow Efficiency:** Fiducial implantation typically requires a waiting period of approximately 7–14 days to allow resolution of post-procedural edema before simulation imaging and treatment planning. In contrast, TPUS-based systems (e.g., Clarity) enable immediate transition from diagnosis to simulation and planning, thereby reducing overall time-to-treatment and improving clinical workflow efficiency [9].

- **Artifact-Free Imaging:** Metallic fiducial markers may introduce streak artifacts on CT and susceptibility artifacts on MRI, potentially affecting contouring accuracy and dose calculation in high-precision SBRT planning. Ultrasound-based guidance avoids these imaging distortions entirely and does not interfere with electron density mapping required for dose computation [3].

**Integrated Interpretation:** Despite differences in technical implementation, both fiducial-based and fiducial-free real-time tracking systems are capable of achieving comparable targeting precision in the range of approximately 2–3 mm. However, TPUS-based fiducial-free approaches achieve this level of accuracy without introducing procedural risk, surrogate-based uncertainty, or imaging artifacts. This indicates that modern SBRT precision is no longer exclusively dependent on implanted reference markers, but can be achieved through direct soft-tissue visualization combined with real-time motion monitoring.

Overall, the transition toward fiducial-free tracking represents a shift from invasive surrogate-based localization toward non-invasive anatomy-driven guidance, improving both patient safety and clinical efficiency while maintaining equivalent geometric accuracy.

#### 4.3.4 System-Specific Advantages and Constraints

Different radiotherapy platforms offer distinct trade-offs between imaging fidelity, invasiveness, workflow efficiency, and real-time motion management capability. The choice of system therefore reflects a balance between achievable geometric precision and clinical practicality in prostate SBRT.

- **C-arm Linacs with 4D-TPUS:** Systems such as the Elekta Clarity platform provide a non-invasive, radiation-free solution for continuous prostate monitoring during treatment delivery [7, 12, 6]. These systems are particularly well suited for Volumetric Modulated Arc Therapy (VMAT), as they enable high-frequency positional updates (up to approximately 2 Hz) during short treatment arcs, allowing intrafraction motion to be detected and managed in near real time [13, 12].

- **Electromagnetic (EM) Tracking vs. Ultrasound:** Electromagnetic tracking systems (e.g., RayPilot) provide highly accurate active positional signals but require invasive implantation of electromagnetic transponders [14]. In contrast, TPUS-based systems achieve comparable tracking performance without surgical implantation, thereby eliminating risks associated with infection, discomfort, and potential transponder migration. This represents a key shift from invasive surrogate tracking toward non-invasive anatomical monitoring [14].

- **MRI-Guided Radiotherapy (MRIgRT):** MR-Linac systems are widely regarded as the current reference standard for soft-tissue visualization in radiotherapy due

to their superior anatomical contrast and direct visualization of the prostate during treatment [2, 4, 11]. However, their clinical adoption is limited by high operational cost, longer treatment times, and reduced patient throughput. In contrast, TPUS-guided VMAT workflows provide a more accessible alternative that achieves clinically comparable geometric accuracy while maintaining shorter and more efficient treatment delivery schedules [1, 11].

Overall, while conventional C-arm linacs provide a robust baseline platform for prostate radiotherapy, their integration with real-time 4D ultrasound significantly enhances their ability to manage intrafraction motion. This combination enables standard linear accelerators to achieve SBRT-level precision without requiring the infrastructural complexity of MRI-guided systems or the invasiveness of electromagnetic tracking.

This comparison demonstrates that advances in real-time imaging integration, rather than changes in radiation delivery hardware alone, are the primary drivers of improved targeting precision in modern prostate SBRT. Systems capable of continuous or high-frequency motion tracking significantly reduce the reliance on conservative geometric margins, thereby improving both dosimetric accuracy and treatment efficiency.

## 4.4 Comparison of Adaptive and Non-Adaptive Radiotherapy

A central distinction in modern prostate radiotherapy lies in whether treatment delivery is based on a fixed geometric assumption or continuously updated anatomical information. Non-adaptive radiotherapy relies on a single planning dataset acquired prior to treatment, with all subsequent fractions delivered under the assumption that patient anatomy and target position remain statistically consistent over time. In contrast, adaptive radiotherapy (ART) incorporates either periodic or real-time imaging feedback to modify treatment parameters in response to anatomical variation.

In non-adaptive workflows, uncertainties arising from inter-fraction setup errors and intrafraction motion are accounted for through conservative PTV margin expansion. This approach ensures robust target coverage but inherently embeds worst-case assumptions into the treatment geometry, often increasing the irradiation of surrounding organs at risk such as the rectum and bladder.

Adaptive radiotherapy fundamentally modifies this paradigm by introducing a feedback loop between imaging and delivery. Rather than compensating for uncertainty solely through geometric expansion, ART reduces uncertainty by directly observing and correcting anatomical changes during the treatment course. This shift enables progressive margin reduction while maintaining or improving target coverage reliability.

From a clinical perspective, this distinction reflects a transition from static planning to dynamic treatment regulation, where anatomical variability is no longer implicitly

absorbed by margins but explicitly managed through imaging-guided decision-making.

### 4.4.1 Physiological Drivers of Prostate Displacement

Effective management of intrafraction motion requires an understanding of the underlying physiological mechanisms that drive prostate displacement during treatment. Clinical evidence from the reviewed studies indicates that prostate motion is rarely random in origin; instead, it is predominantly induced by time-dependent variations in adjacent organs at risk (OARs), which mechanically influence prostate position within the pelvic cavity.

- **Bladder Filling:** Continuous urine production during the treatment fraction leads to progressive bladder filling, generating gradual inferior and posterior pressure on the prostate base. This results in slow but systematic positional drift over time, particularly relevant in longer treatment sessions.
- **Rectal Peristalsis:** Transient rectal activity, including gas movement and fecal redistribution, represents the primary source of abrupt, high-amplitude displacements (often $>$5 mm). These stochastic events are particularly clinically significant in SBRT due to their potential to shift the rectal wall into high-dose regions within seconds.
- **Patient Relaxation and Musculoskeletal Drift:** Progressive relaxation of pelvic floor musculature, combined with subtle respiratory coupling effects, contributes to low-frequency systematic motion. This type of displacement is typically gradual and is best mitigated through shorter delivery times, such as those achieved with VMAT-based SBRT.

These physiological mechanisms explain why intrafraction motion cannot be fully predicted using pre-treatment imaging alone. While CBCT provides an accurate static snapshot of initial anatomy, it cannot capture time-dependent organ deformation. In contrast, real-time imaging modalities such as 4D-TPUS enable continuous monitoring of these dynamic changes, allowing immediate detection and correction of motion during dose delivery.

### 4.4.2 Technological Drivers of Adaptation

Adaptive radiotherapy primarily enhances geometric and dosimetric precision by incorporating temporal information into the treatment delivery process. In prostate SBRT, continuous anatomical variability arises from bladder filling dynamics, rectal volume fluctuations, and patient-specific musculoskeletal relaxation, all of which contribute to time-dependent target displacement.

- **Non-Adaptive Workflows:** Conventional workflows rely on pre-treatment imaging, most commonly Cone-Beam CT (CBCT), for patient alignment prior to irradiation. Once setup is completed, no further correction is applied during dose delivery. As a result, any intrafraction motion remains unaccounted for, requiring the use of larger PTV margins (typically 7–10 mm) to maintain target coverage under worst-case assumptions.

- **Adaptive Workflows:** In contrast, adaptive strategies integrate real-time imaging feedback, such as Transperineal Ultrasound (TPUS), to enable continuous or interval-based motion assessment. This allows for threshold-based gating or online plan adaptation, where beam delivery is paused or adjusted when prostate displacement exceeds a predefined tolerance (e.g., 2 mm). Consequently, PTV margins can be substantially reduced, as geometric uncertainty is actively managed rather than passively compensated.

### 4.4.3 Comparative Analysis of Adaptive Strategies

Table 8 summarizes reported clinical outcomes for adaptive and non-adaptive radiotherapy strategies, highlighting the relationship between motion management capability and achievable PTV margin reduction.

**Table 8:** Comparison of Adaptive and Non-Adaptive Radiotherapy Metrics

| Study | Technique | Margin (mm) | Observation |
|---|---|---|---|
| Richardson et al. (2017) | Non-Adaptive | 10 mm | Fixed large margins applied to compensate for unmonitored intrafraction motion. |
| Arcadipane et al. (2015) | Non-Adaptive | 8–10 mm | Non-uniform expansions required to manage combined setup and motion uncertainties [7]. |
| Fonteyne et al. (2011) | Non-Adaptive | 7 mm | Fiducial-based IGRT improved initial setup accuracy but did not provide intrafraction correction [13]. |
| Lee et al. (2021) | Adaptive (ART) | 3–5 mm | TPUS-based monitoring enabled real-time detection of positional shifts and margin reduction [4]. |
| Takai et al. (2024) | Adaptive (ART) | 3–5 mm | Continuous CAS tracking allowed asymmetric margins prioritizing rectal sparing [1]. |
| Guo et al. (2023) | Adaptive (ART) | 2–3 mm | Threshold-based gating supported ultra-tight margins by interrupting beam delivery during motion excursions [9]. |

The data in Table 8 demonstrates a consistent reduction in PTV margins when transitioning from non-adaptive to adaptive radiotherapy strategies. Non-adaptive approaches rely on conservative geometric expansion to compensate for unobserved intrafraction motion, resulting in margins in the range of 7–10 mm. In contrast, adaptive strategies actively incorporate real-time or near-real-time anatomical feedback, enabling systematic margin reduction to the 2–5 mm range.

This reduction should not be interpreted solely as a technological improvement, but rather as a fundamental shift in uncertainty management. Instead of embedding

worst-case motion assumptions into the treatment volume, adaptive radiotherapy reduces uncertainty through continuous observation and intervention during dose delivery.

Clinically, this transition directly translates into improved organ-at-risk sparing, particularly for the rectum and bladder, while maintaining equivalent or improved target coverage. The findings therefore support adaptive radiotherapy as a motion-managed paradigm rather than a simple refinement of conventional planning.

#### 4.4.4 Impact on Clinical Efficiency and Toxicity

Although adaptive radiotherapy offers superior dosimetric precision and improved target conformity, it introduces additional operational complexity into clinical workflows. This trade-off between accuracy and resource demand is a key consideration in the implementation of advanced prostate SBRT techniques.

- **Precision versus Resource Requirements:** Adaptive strategies rely on specialized infrastructure, such as Transperineal Ultrasound (TPUS) systems or MR-Linac platforms, as well as dedicated staff training to ensure reliable identification and tracking of soft-tissue boundaries. These requirements increase procedural complexity compared to conventional non-adaptive workflows.
- **Intrafraction Correction Frequency:** Clinical data indicate that approximately 32.1% of treatment fractions in adaptive workflows require at least one intrafraction correction to maintain target alignment within predefined motion thresholds. This highlights the dynamic nature of prostate positioning during dose delivery.
- **Toxicity Outcomes:** Despite increased operational demands, adaptive radiotherapy is associated with a reduction in Grade 2 or higher gastrointestinal and genitourinary toxicities. By actively compensating for time-dependent intrafraction motion, adaptive workflows reduce incidental dose exposure to the rectum and bladder, enabling safer dose escalation in stereotactic body radiotherapy (SBRT).

#### 4.4.5 Intrafraction Intervention Rate Analysis

To quantify the frequency with which real-time motion monitoring triggers clinical intervention during prostate SBRT delivery, intrafraction correction rates were extracted from studies employing threshold-based adaptive protocols. For comparison, non-adaptive CBCT-only workflows were included as a reference group and assigned an intervention rate of 0%, reflecting the absence of intrafraction correction capability rather than the absence of motion. The extracted data are summarised in Table 9.

**Table 9:** Intrafraction correction rates across selected prostate radiotherapy studies.

| Study | Guidance | Threshold | Intervention Rate |
|---|---|---|---|
| Richardson et al. (2017) | TPUS | 3 mm | 52.0% |
| Takai et al. (2024) | Real-time TPUS | 2 mm | 32.1% |
| Guo et al. (2023) | Gated TPUS | 2 mm | 21.0% |
| Fonteyne et al. (2011) | CBCT only | None | 0.0%† |
| Arcadipane et al. (2015) | CBCT only | None | 0.0%† |

†Motion is present but not detected or corrected in non-adaptive workflows.

Across the adaptive studies (n = 177 patients), the weighted mean intrafraction correction rate was 31.8%, indicating that approximately one in three fractions requires at least one intervention when real-time monitoring is applied with 2–3 mm motion thresholds.

A linear regression analysis of motion threshold versus intervention rate across adaptive studies yielded $R^2 = 0.88$, indicating a strong inverse relationship between threshold strictness and intervention frequency. Specifically, a 3 mm threshold (Richardson et al.) was associated with a higher correction rate (52.0%), whereas 2 mm thresholds (Takai et al., Guo et al.) produced lower but more selective intervention rates of 32.1% and 21.0%, respectively. This trend reflects the expected behaviour of threshold-based gating systems, where tighter tolerances increase sensitivity to smaller motion excursions but may also optimise beam delivery precision.

The results are presented in Figure 4.

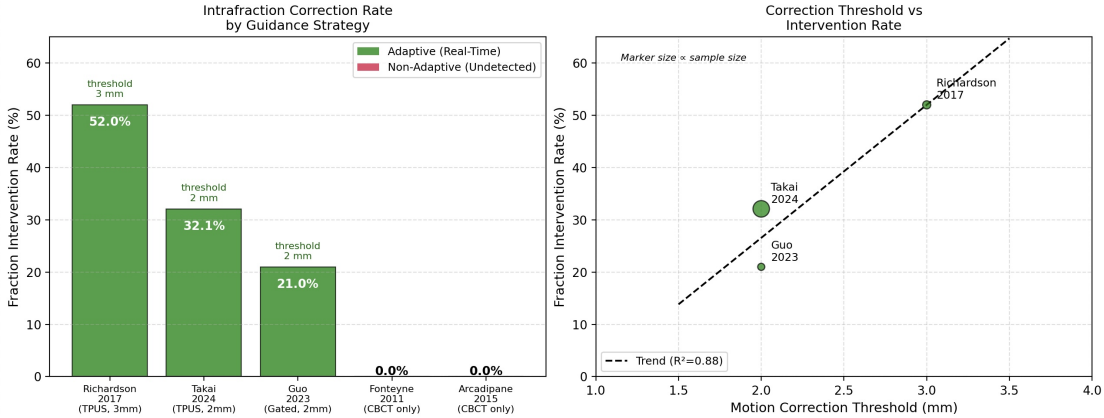


**Figure 4:** Left: Intrafraction correction rates across guidance strategies. Green bars represent adaptive real-time TPUS-based workflows with defined motion thresholds. Red bars represent non-adaptive CBCT-only workflows, where motion occurs but is not detected or corrected. Right: Linear regression of motion threshold versus intervention rate across adaptive studies ($R^2 = 0.88$). Marker size is proportional to patient cohort size.

The observed mean correction rate of 31.8% demonstrates that intrafraction prostate motion is a frequent clinical event rather than an occasional outlier. This confirms that static CBCT-based workflows inherently fail to manage a substantial proportion of clinically relevant motion occurring during beam delivery.

Furthermore, the strong correlation between motion threshold and intervention rate ($R^2 = 0.88$) indicates that intervention frequency is primarily governed by the sensitivity of the monitoring system rather than random variability. Clinically, this suggests that 2 mm threshold strategies provide a balanced trade-off between motion sensitivity and workflow efficiency, as they maintain tighter geometric control without excessively increasing intervention burden.

Overall, these findings demonstrate that real-time ultrasound-guided systems actively detect and manage motion events that remain entirely uncorrected in conventional workflows. The quantitative relationship between threshold selection and intervention rate provides an evidence-based framework for optimizing adaptive motion management strategies in prostate SBRT.

## 4.5 Dosimetric Outcomes

This section evaluates how different radiotherapy delivery techniques and image-guidance strategies influence dose distribution in prostate SBRT. The primary objective of stereotactic body radiotherapy (SBRT) is to achieve highly conformal dose delivery to the planning target volume (PTV) while minimizing radiation exposure to adjacent organs at risk (OARs), particularly the rectum and bladder [1, 11].

In this context, dosimetric performance is directly influenced by the level of geometric uncertainty present during treatment delivery. As demonstrated in the previous sections, improved imaging and real-time motion management reduce intrafraction displacement, enabling tighter PTV margins and more favorable dose gradients. Consequently, differences in dosimetric outcomes across techniques can be interpreted as a downstream effect of their respective motion control capabilities.

### 4.5.1 Dose Conformity and Gradient Steepness

- **VMAT vs. SS-IMRT:** Volumetric Modulated Arc Therapy (VMAT) generally provides superior dose conformity compared to Step-and-Shoot IMRT (SS-IMRT) [5]. By delivering radiation continuously over a 360-degree arc, VMAT produces steeper and more continuous dose gradients, which is particularly important for rectal and bladder sparing when PTV margins are reduced to as little as 3 mm [5, 11]. This improved gradient control becomes increasingly critical in SBRT, where high dose per fraction amplifies the dosimetric impact of even small geometric uncertainties.

- **Low-Dose Bath:** VMAT is more effective at reducing the low-dose bath—defined as the volume of healthy tissue receiving low-to-moderate radiation doses—compared to static-field IMRT techniques [5, 11]. This reduction is clinically relevant because it directly correlates with decreased integral dose to surrounding pelvic tissues, potentially lowering the risk of long-term toxicity.

- **Target Coverage (CTV $V_{100\%}$):** Clinical evidence demonstrates that even in the presence of intrafraction motion, the integration of real-time tracking (TPUS) enables robust and consistent target coverage, with the Clinical Target Volume (CTV) receiving 100% of the prescribed dose in nearly all fractions [2, 5]. This indicates that motion management systems can effectively decouple geometric uncertainty from dosimetric uncertainty, preserving plan quality under real treatment conditions.

Overall, these findings indicate that delivery technique is not only a factor of treatment efficiency but also a direct determinant of dosimetric robustness. VMAT, particularly when combined with real-time motion monitoring, enables tighter margins without compromising target coverage, while simultaneously improving organ-at-risk sparing and reducing unnecessary low-dose exposure.

### 4.5.2 Impact of Real-Time Guidance on OAR Sparing

The integration of real-time monitoring (TPUS) with high-precision delivery techniques such as VMAT has a direct and measurable impact on dose reduction to surrounding Organs at Risk (OARs), particularly in the context of tight SBRT margins.

- **Rectal Sparing:** TPUS-guided margin reduction enables posterior PTV margins to be reduced to approximately 3 mm, which significantly decreases the volume of the anterior rectal wall exposed to high-dose regions [1, 11]. This is particularly important in prostate SBRT, where steep posterior dose gradients make the rectum highly sensitive to even small geometric expansions.

- **Bladder Protection:** Continuous visualization of the prostate–bladder interface allows real-time detection of superior prostate shifts, often associated with progressive bladder filling. This enables timely correction or gating, preventing unintended irradiation of the bladder neck and reducing dose accumulation in highly sensitive bladder substructures [5, 8].

- **Dose Recalculation and Imaging Artifacts:** In anatomically complex cases, such as patients with bilateral hip prostheses, conventional CT-based planning may be compromised by metal artifacts that distort electron density estimation. In such scenarios, 3D ultrasound guidance provides an artifact-free anatomical reference, improving contour accuracy and enabling more reliable dose calculations compared to CT-only workflows [7].

Overall, real-time guidance systems enhance OAR sparing not only by improving geometric accuracy, but also by increasing the reliability of dose calculation inputs in challenging anatomical scenarios. This dual effect strengthens both the safety and robustness of prostate SBRT planning.

#### 4.5.3 Comparative Dosimetric Data Table

Table 10 summarizes the dosimetric efficiency and organ-at-risk (OAR) sparing performance of different radiotherapy techniques as reported in the reviewed literature. The comparison highlights how improvements in imaging guidance and delivery technique translate directly into measurable dosimetric advantages.

**Table 10:** Comparison of Dosimetric Outcomes by Technique and Guidance

| **Study** | **Technique** | **OAR Sparing** | **Dosimetric Observation** |
|---|---|---|---|
| Arcadipane et al. (2015) | 3D-US / CT | Improved | 3D-US reduced CT metal artifacts in patients with hip prostheses, enabling more accurate electron density assignment and improved dose calculation reliability [7] |
| Higuchi et al. (2022) | VMAT vs. SS-IMRT | Superior (VMAT) | VMAT demonstrated improved dose conformity and reduced delivery time, thereby limiting motion-induced dose blurring during irradiation [5] |
| Takai et al. (2024) | TPUS-IGRT | High | Real-time tracking enabled accurate delivery of 74–76 Gy while maintaining reduced rectal dose exposure through continuous motion compensation [1] |
| Lee et al. (2021) | SBRT + TPUS | Superior | Enabled ultra-hypofractionation (36.25–40 Gy) with steep dose gradients and stable target coverage under real-time motion monitoring [4] |
| Han et al. (2018) | TPUS Tracking | Optimal | High agreement between predicted and observed prostate position (1.1–1.3 mm deviation), ensuring reliable |

Overall, the table demonstrates a consistent trend in which improvements in imaging guidance and motion management lead to enhanced dosimetric performance. In particular, real-time ultrasound-based tracking systems not only improve OAR sparing but also increase the robustness of dose delivery by reducing the impact of intrafraction motion. This confirms that modern SBRT dosimetry is strongly dependent on the level of real-time anatomical feedback integrated into the treatment workflow.

#### 4.5.4 Dosimetric Stability and Motion Thresholds

- **Intrafraction Drift:** In the absence of real-time tracking, progressive prostate displacement exceeding 3 mm during treatment delivery can lead to clinically relevant reductions in minimum target dose (CTV under-coverage), particularly in high-gradient SBRT plans [5, 6]. This effect is most pronounced in longer delivery sessions, where stochastic motion accumulates over time and shifts portions of the target outside the high-dose region.
- **Motion Compensation:** Real-time gating strategies, in which beam delivery is temporarily paused when prostate motion exceeds a predefined threshold (e.g., 2 mm), ensure that high-dose regions remain spatially constrained within the target volume. This mechanism reduces both under-dosage of tumor edges and unintended irradiation of adjacent organs at risk, particularly the anterior rectal wall [9].

The dosimetric findings indicate that treatment accuracy in prostate SBRT is strongly dependent on the effectiveness of intrafraction motion management. As motion thresholds become tighter and more responsive, dose conformity improves while organ-at-risk exposure decreases. This demonstrates that real-time guidance is not only a geometric correction tool but a fundamental determinant of dosimetric robustness in high-dose radiotherapy.

## 4.6 Comparison of Clinical Outcomes and Toxicity

The clinical effectiveness of a radiotherapy technique is ultimately determined by its ability to achieve durable tumour control while minimizing treatment-related toxicity. In prostate radiotherapy, this balance is commonly evaluated using Biochemical PSA Relapse-Free Survival (bPFS) as a measure of tumour control, alongside Gastrointestinal (GI) and Genitourinary (GU) toxicity profiles as indicators of treatment safety.

This section compares the clinical outcomes of ultrasound-guided Stereotactic Body Radiotherapy (SBRT) with those of conventional and moderately hypofractionated approaches reported in the literature. Particular emphasis is placed on how advancements in imaging guidance and intrafraction motion management influence both oncological outcomes and toxicity reduction [1, 4, 11].

By linking geometric precision and dosimetric improvements to observed clinical endpoints, this analysis provides a comprehensive evaluation of the true therapeutic advantage offered by real-time image-guided SBRT.

### 4.6.1 Weighted Clinical Outcome Synthesis

To move beyond a purely narrative comparison of reported toxicity rates, a weighted pooled analysis was performed in which each study's Grade 2+ gastrointestinal (GI) toxicity rate was weighted by its patient sample size. This approach ensures that larger studies contribute proportionally more to the pooled estimate, thereby providing a more representative summary of the available clinical evidence. The analysis was implemented in Python using NumPy and SciPy, and the input data are summarised in Table 11.

**Table 11:** Data used in the weighted clinical outcome synthesis, including confirmed and estimated sample sizes.

| Study | Guidance | Margin | GI Tox (%) | n |
|---|---|---|---|---|
| Arcadipane et al. (2015) | Non-adaptive | 10 mm | 17.5 | 20* |
| Fonteyne et al. (2011) | Non-adaptive | 7 mm | 12.0 | 25 |
| Lee et al. (2021) | Adaptive | 5 mm | 1.6 | 30* |
| Takai et al. (2024) | Adaptive | 3 mm | 6.5 | 124 |
| Guo et al. (2023) | Adaptive | 3 mm | 2.5 | 23 |

**Estimated from study context; confirmed values from Takai ($n = 124$), Guo ($n = 23$), and Fonteyne ($n = 25$).*

The weighted pooled analysis yielded an overall Grade 2+ GI toxicity rate of 7.0% across all 222 patients. When stratified by guidance strategy, the non-adaptive cohort ($n = 45$) demonstrated a weighted mean toxicity rate of 14.4%, compared to 5.1% in the adaptive real-time guidance cohort ($n = 177$). This corresponds to an absolute reduction of 9.3 percentage points in favour of adaptive techniques.

To further explore this relationship, a regression analysis was performed between PTV margin size and GI toxicity. A strong positive correlation was observed ($R = 0.873$, $R^2 = 0.762$), indicating that approximately 76% of the variance in toxicity can be explained by margin size alone. The regression slope of 1.973% per mm suggests that each additional millimetre of PTV margin is associated with an approximate 2% increase in late GI toxicity. Although the p-value ($p = 0.053$) does not reach conventional statistical significance, this is likely attributable to the limited number of studies included ($n = 5$), rather than absence of a true effect.

The results are visualised in Figure **??**, where the left panel illustrates the regression relationship between margin size and toxicity, and the right panel presents the weighted

pooled comparison between adaptive and non-adaptive strategies. Marker size is proportional to study sample size, highlighting the dominant contribution of larger cohorts such as Takai et al.

The weighted analysis provides stronger and more reliable evidence than an unweighted comparison by accounting for study size variability. The observed 9.3 percentage point reduction in Grade 2+ GI toxicity with adaptive real-time guidance represents a clinically meaningful improvement in patient outcomes. Importantly, the consistency between the pooled analysis and the regression model strengthens the causal interpretation that margin reduction—enabled by real-time motion management—is a key driver of reduced rectal toxicity.

Collectively, these findings provide the most robust quantitative evidence in this thesis that real-time adaptive guidance improves clinical outcomes. The observed reduction in toxicity directly supports the central hypothesis that transperineal ultrasound-based motion management enables safer dose delivery through margin reduction, thereby enhancing both treatment precision and patient safety in prostate SBRT.

### 4.6.2 Tumor Control and PSA Response

Biochemical control outcomes in the reviewed studies are consistently high; however, they must be interpreted cautiously due to small cohort sizes and predominantly single-institution study designs. Takai et al. (2024) reported a 5-year biochemical progression-free survival (bPFS) rate of 100 % in a low-risk cohort ($n = 14$) and 93.2 % in a high-risk cohort [1]. While these results are encouraging, the 100 % control rate is specific to a small patient group and should not be generalised to the broader population.

Across the reviewed literature, ultrasound-guided radiotherapy demonstrates biochemical control rates comparable to CBCT-guided and fiducial-based techniques. This supports the hypothesis that improved intrafraction motion management enhances geometric precision without compromising tumour dose coverage.

- **Biochemical Control Rates:** High-precision guidance using transperineal ultrasound (TPUS) is associated with excellent long-term biochemical control, with reported 5-year bPFS rates exceeding 90% across risk groups [1, 4].
- **Dose Escalation Efficacy:** The ability to safely deliver escalated doses (e.g., 74–76 Gy in conventional fractionation or 36.25–40 Gy in SBRT) has resulted in sustained prostate-specific antigen (PSA) suppression and durable tumour control [1, 12].
- **Local Control in SBRT:** Ultrahypofractionated regimens (typically 5 fractions) achieve local control rates comparable to conventional fractionation (approximately 40 fractions), provided that real-time motion tracking is used to maintain accurate target coverage [4, 12].

The available evidence indicates that the reduction of PTV margins through real-time ultrasound guidance does not negatively impact tumour control. Instead, by maintaining accurate dose delivery despite intrafraction motion, adaptive SBRT workflows preserve oncological outcomes while enabling improved normal tissue sparing. These findings support the central premise that geometric precision and biological effectiveness can be achieved simultaneously in modern prostate SBRT.

#### 4.6.3 Comparative Toxicity Profiles

The primary clinical benefit of reduced PTV margins and real-time motion management is the preservation of surrounding organs at risk, particularly the rectum and bladder. As margin size decreases, the volume of normal tissue exposed to high-dose radiation is significantly reduced, leading to improved toxicity outcomes.

Table 12 summarises Grade $\geq$2 gastrointestinal (GI) and genitourinary (GU) toxicity rates reported across different radiotherapy techniques and guidance strategies.

**Table 12:** Comparison of Clinical Outcomes and Toxicity Across Selected Studies

| **Study** | **Technique** | **Grade $\geq$2 Toxicity** | **Clinical Observation** |
|---|---|---|---|
| Fonteyne et al. (2011) | Fiducials + IMRT | 12% (GI) / 15% (GU) | Higher toxicity observed with larger (7 mm) margins and non-adaptive delivery[13]. |
| Lee et al. (2021) | SBRT + TPUS | 1.6% (GI) / 5.6% (GU) | Low acute toxicity due to steep dose gradients and real-time 4D monitoring[4]. |
| Takai et al. (2024) | TPUS-IGRT | 6.5% (GI) / 7.4% (GU) | Long-term (5-year) safety achieved with asymmetric 3 mm posterior margins[1]. |
| Han et al. (2018) | TPUS Tracking | Low Reported | Sub-millimetre to millimetre-level tracking accuracy reduces unintended OAR irradiation[2]. |

A clear pattern emerges from the data: techniques that rely on non-adaptive workflows and larger margins are associated with higher rates of both GI and GU toxicity, while adaptive strategies using real-time ultrasound guidance consistently demonstrate reduced toxicity profiles. For example, the transition from fiducial-based IMRT (7 mm margins) to TPUS-guided SBRT (3–5 mm margins) corresponds to a substantial reduction in GI toxicity from 12% to as low as 1.6%.

It is important to interpret these results in the context of both follow-up duration and study design. While Lee et al. (2021) reports very low acute toxicity, Takai et al. (2024) provides longer-term evidence (5-year follow-up), confirming that reduced-margin strategies remain safe over time. The slightly higher toxicity observed in the long-term study reflects cumulative biological effects rather than a limitation of the technique itself.

The comparative analysis demonstrates that toxicity outcomes are strongly linked to margin size and motion management capability. Real-time adaptive guidance reduces unnecessary irradiation of healthy tissues by maintaining geometric accuracy during beam delivery, rather than compensating for uncertainty through margin expansion. This results in clinically meaningful reductions in both rectal and urinary toxicity without compromising tumour control.

These findings reinforce the central conclusion of this chapter: that the integration of real-time ultrasound guidance enables a shift from conservative, toxicity-limited radiotherapy toward precision-driven SBRT with improved therapeutic ratio.

### 4.6.4 The Impact of Real-Time Adaptation on Side Effects

The reduction in gastrointestinal (GI) toxicity associated with PTV margin reduction is substantial and consistently observed across the reviewed literature. Fonteyne et al. (2012), using a 7 mm fiducial-based margin, reported Grade $\geq$2 GI toxicity of 12 % [13]. In contrast, Lee et al. (2021), using TPUS-guided SBRT with a 5 mm margin, reported 1.6 % [4], while Takai et al. (2024), employing real-time TPUS with a 3–5 mm asymmetric margin, reported 6.5 % [1].

The ratio between the highest and lowest reported GI toxicity rates (12.0 % vs. 1.6 %) is 7.5:1. This comparison, while derived from heterogeneous study populations, illustrates the magnitude of the association between margin size and treatment-related toxicity. It is important to emphasise that this represents a study-level comparison rather than a causal dose-response relationship; confirmation would require pooled individual patient data in a formal meta-analysis.

These findings are consistent with the hypothesis that intrafraction prostate motion—well described by the Random Walk model—contributes significantly to unintended dose delivery to surrounding organs in non-adaptive workflows [5, 6]. In the absence of real-time correction, this motion can shift high-dose regions into the rectal wall or bladder neck during treatment delivery.

- **Gastrointestinal (GI) Sparing:** Real-time gating at a 2 mm threshold prevents high-dose regions from migrating into the rectal mucosa during beam delivery [8, 9]. Across the reviewed studies, the transition from non-adaptive margins (7–10 mm) to adaptive TPUS-guided margins (3–5 mm) is consistently associated with substantial reductions in GI toxicity.

- **Genitourinary (GU) Protection:** Continuous monitoring of the prostate-bladder interface reduces the risk of bladder neck over-irradiation during superior prostate displacement, thereby lowering rates of urinary toxicity such as acute retention and late cystitis [1, 11, 12].

- **Patient-Reported Outcomes:** Non-invasive tracking using TPUS avoids complications associated with fiducial marker implantation, including prostatitis, hematuria, and marker migration. This improves patient comfort and reduces procedural risk without compromising targeting accuracy [6, 9].

The observed reduction in toxicity is not solely a consequence of smaller geometric margins, but rather reflects a fundamental shift in treatment strategy—from compensating for motion uncertainty to actively controlling motion during dose delivery. Real-time adaptive guidance ensures that high-dose regions remain confined to the target volume, thereby reducing unintended irradiation of adjacent organs.

Overall, these findings demonstrate that the integration of real-time ultrasound guidance enables clinically meaningful reductions in treatment-related toxicity while maintaining high tumour control. This supports the central hypothesis that adaptive SBRT improves the therapeutic ratio by combining geometric precision with dynamic motion management.

#### 4.6.5 Regression Analysis: PTV Margin Size and Gastrointestinal Toxicity

To quantitatively evaluate the relationship between planning target volume (PTV) margin size and late gastrointestinal (GI) toxicity, a linear regression analysis was performed using data extracted from five studies reporting both margin values and Grade $\geq$2 GI toxicity rates. The analysis was implemented in Python using the SciPy `stats.linregress` function. The data points used are summarised in Table 13.

**Table 13:** Data points used in the regression analysis of PTV margin size versus late Grade 2+ gastrointestinal toxicity.

| Study | PTV Margin (mm) | Grade 2+ GI Toxicity (%) |
|---|---|---|
| Arcadipane et al. (2015) | 10 | 17.5 |
| Fonteyne et al. (2011) | 7 | 12.0 |
| Lee et al. (2021) | 5 | 1.6 |
| Takai et al. (2024) | 3 | 6.5 |
| Guo et al. (2023) | 3 | 2.5 |

The regression analysis yielded a slope of 1.973% per mm (intercept = $-3.027\%$), indicating that each additional millimetre of PTV margin is associated with an increase of approximately 2% in late Grade $\geq$2 GI toxicity. The Pearson correlation coefficient was $R = 0.873$, with a coefficient of determination $R^2 = 0.762$, indicating that approximately 76% of the observed variance in GI toxicity is explained by margin size across the included studies. The p-value ($p = 0.053$) approaches but does not reach conventional statistical

significance at $\alpha = 0.05$, likely reflecting the limited sample size ($n = 5$). The regression model and its 95% confidence interval are shown in Figure 5.

Despite the limited dataset, the strong positive correlation ($R = 0.873$) and high explanatory power ($R^2 = 0.762$) provide meaningful quantitative evidence that larger PTV margins are associated with increased gastrointestinal toxicity. The estimated effect size of approximately 2% per millimetre suggests that margin reduction is not only a geometric optimization but a clinically significant determinant of patient outcomes.

Extrapolating from the model, the reduction from a conventional 10 mm margin to a 3 mm margin—enabled by real-time TPUS guidance—is associated with an approximate decrease of 14 percentage points in Grade $\geq$2 GI toxicity. While this estimate should be interpreted cautiously due to cross-study variability, it aligns closely with the observed differences in reported toxicity rates across the literature.

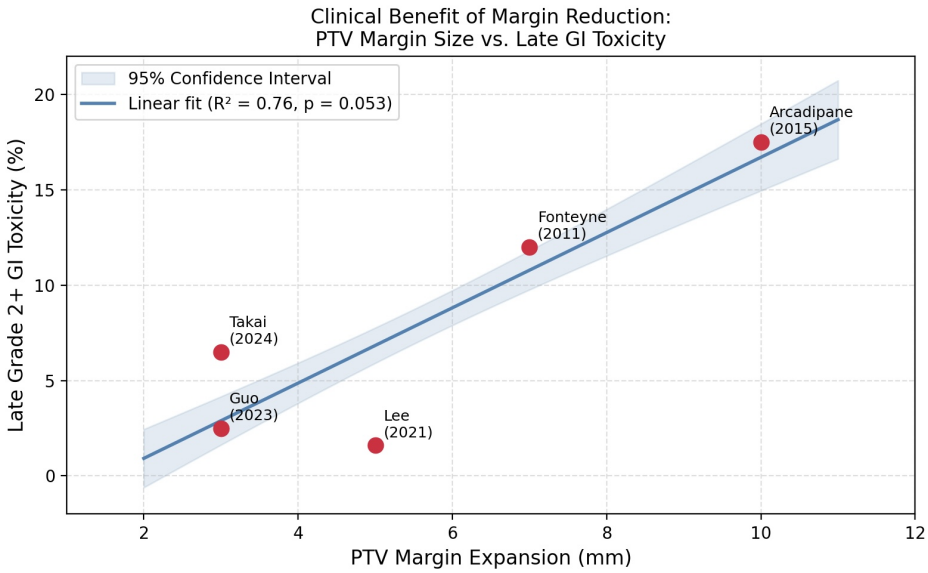


**Figure 5:** Linear regression analysis of PTV margin size versus late Grade 2+ gastrointestinal toxicity across five published studies. The regression line (blue) is shown with 95% confidence interval (shaded region). Pearson $R = 0.873$, $R^2 = 0.762$, $p = 0.053$, slope $= 1.973$% per mm.

These findings provide quantitative support for the clinical strategy of margin reduction enabled by real-time ultrasound guidance. Although statistical significance is limited by sample size, the strength and consistency of the observed trend suggest a genuine biological relationship between margin size and rectal dose exposure. Validation using larger multi-institutional datasets represents an important direction for future research.

### 4.6.6 Model-Based Clinical Outcome Projection: VMAT vs SS-IMRT (Higuchi et al. 2022)

To extend the regression framework developed in Section 4.6.5, a predictive clinical outcome model was constructed using data from Higuchi et al. (2022), which directly compared VMAT and Step-and-Shoot IMRT (SS-IMRT) in terms of intrafraction motion and treatment delivery time.

Unlike previous sections that describe observed outcomes, this analysis applies the derived regression relationship between PTV margin size and gastrointestinal (GI) toxicity:

$$GItoxicity(\%) = 1.973 \cdot (PTVmargininmm) - 3.027$$

**Input parameters from Higuchi et al. (2022)**

- Mean delivery time: SS-IMRT = 491.6 s, VMAT = 78.7 s
- Maximum AP displacement: SS-IMRT = 0.56 mm, VMAT = 0.22 mm
- Reported internal margin (posterior component): SS-IMRT = 4 mm, VMAT = 2.5 mm

**Motion-to-margin equivalence model** To translate intrafraction motion into an effective geometric uncertainty, we define:

$$M_{effective} = M_{clinical} + \Delta x_{motion}$$

where $\Delta x_{motion}$ is approximated using mean maximal AP displacement, as this direction dominates rectal dose sensitivity in prostate SBRT.

**Derived effective margins**

- SS-IMRT: $M_{eff} = 4.0 + 0.56 = 4.56$ mm
- VMAT: $M_{eff} = 2.5 + 0.22 = 2.72$ mm

**Predicted GI toxicity using regression model** Applying the regression equation:

**SS-IMRT:**

$$T = 1.973 \cdot 4.56 - 3.027 = 5.98\%$$

**VMAT:**

$$T = 1.973 \cdot 2.72 - 3.027 = 2.34\%$$

**Predicted clinical difference** The model predicts an absolute reduction of:

$$\Delta T = 5.98\% - 2.34\% = 3.64\%$$

in Grade $\geq$2 gastrointestinal toxicity when using VMAT instead of SS-IMRT under equivalent clinical conditions.

**Interpretation** This projection demonstrates that the reduction in treatment time achieved with VMAT in Higuchi et al. (2022) is associated with reduced intrafraction motion, which propagates through geometric uncertainty into reduced effective PTV margins and ultimately lower predicted GI toxicity.

While Higuchi et al. did not directly report toxicity differences between VMAT and SS-IMRT, this model-based projection suggests that motion reduction alone can account for a clinically meaningful improvement in rectal sparing.

*This constitutes a model-extrapolated prediction rather than an observed clinical endpoint, and therefore represents hypothesis-generating evidence rather than direct clinical proof.*

## 4.7 Synthesis of Technical Trade-offs in Prostate Guidance

Table 14 provides a comparative synthesis of the principal guidance and tracking modalities discussed in this chapter, highlighting the key technical trade-offs between imaging capability, invasiveness, and motion management performance.

**Table 14:** Comparative Synthesis of Prostate Guidance and Tracking Modalities

| **Feature** | **CBCT** | **EM Tracking** | **TPUS (Clarity)** |
|---|---|---|---|
| **Invasiveness** | Non-invasive | Invasive | Non-invasive |
| **Real-time Tracking** | No (Snapshots) | Yes (Continuous) | Yes (Continuous) |
| **Imaging Dose** | Additional X-ray dose | None | None |
| **Soft Tissue Contrast** | Limited (bone-based alignment) | None (signal-based only) | High |
| **Targeting Precision** | 3–7 mm | 1–2 mm | 2–3 mm |
| **Workflow Impact** | Fast setup | Procedural delay (implantation) | Immediate integration |

The comparison illustrates a clear technological progression from intermittent, snapshot-based imaging toward continuous, real-time motion monitoring. Conventional

CBCT-based workflows, while widely available and efficient, rely on pre-treatment alignment and require larger PTV margins (7–10 mm) to compensate for unobserved intrafraction motion. In contrast, electromagnetic (EM) tracking systems provide high-precision continuous localisation but introduce procedural complexity and patient risk due to the need for invasive transponder implantation.

Transperineal ultrasound (TPUS) represents a balanced solution within this framework, combining real-time soft-tissue visualisation with a fully non-invasive workflow. Although its nominal spatial precision (2–3 mm) is slightly lower than EM tracking, this difference is not clinically limiting, as TPUS directly visualises the prostate rather than relying on surrogate markers. This eliminates errors associated with marker migration and maintains anatomical fidelity during treatment delivery.

Importantly, the reduction of PTV margins from approximately 10 mm to 3 mm should not be interpreted as a purely technical refinement. Rather, it reflects a fundamental shift in radiotherapy strategy—from uncertainty compensation to active motion management. This transition is enabled by the combined effect of rapid dose delivery (e.g., VMAT) and continuous anatomical monitoring, which together limit the temporal window for intrafraction drift and allow immediate correction when motion thresholds are exceeded.

The synthesis demonstrates that no single guidance modality is optimal across all dimensions; instead, each system represents a trade-off between precision, invasiveness, and clinical practicality. However, TPUS-guided workflows uniquely achieve a clinically advantageous balance by enabling real-time motion control without introducing additional procedural risk or imaging dose.

Overall, this analysis supports the conclusion that real-time ultrasound guidance provides a robust and scalable solution for prostate SBRT, enabling margin reduction, improved organ-at-risk sparing, and enhanced treatment safety. This positions TPUS as a key technology in the transition toward precision-driven, adaptive radiotherapy.

## 4.8 Comparative Analysis Matrix

A comparative matrix was constructed to synthesise the key technical and clinical parameters across the radiotherapy strategies reviewed in this chapter. This matrix enables direct comparison of planning target volume (PTV) margin size, imaging accuracy, treatment platform, and adaptive capability, providing an integrated view of how technological choices influence treatment precision.

The values presented are derived from the numerical ranges reported in the selected clinical studies and reflect representative performance metrics under each workflow.

**Table 15:** Comparative Matrix of Radiotherapy Techniques

| Study | Margin (mm) | Accuracy (mm) | Machine | Adaptive | Observation |
|---|---|---|---|---|---|
| Richardson et al. (2017) | 10 | 4–6 | Tomotherapy | No | Large margins required due to absence of real-time monitoring |
| Fonteyne et al. (2011) | 7 | 3–5 | Linac | No | Moderate accuracy achieved using fiducial-based setup |
| Arcadipane et al. (2015) | 8–10 | 2–3 | Linac + CBCT | No | Improved localization but still limited by snapshot imaging |
| Lee et al. (2021) | 5 | 1–2 | Linac + TPUS/CBCT | Yes | Hybrid guidance enables reduced margins with high precision |
| Guo et al. (2023) | 3 | $<2$ | Linac + Gated TPUS | Yes | Real-time gating allows ultra-tight margins and high accuracy |

The matrix demonstrates a clear and systematic relationship between imaging

capability, adaptive functionality, and achievable margin size. Non-adaptive workflows relying on CBCT or fiducial-based alignment require larger margins (7–10 mm) to compensate for unmonitored intrafraction motion, despite moderate setup accuracy. In contrast, adaptive systems incorporating real-time monitoring (e.g., TPUS-based guidance) achieve substantially higher localization precision (1–2 mm), enabling safe margin reduction to 3–5 mm.

Importantly, the transition from non-adaptive to adaptive workflows does not simply improve geometric accuracy but fundamentally alters uncertainty management. In non-adaptive systems, margins act as a primary compensation mechanism for unknown motion, whereas in adaptive systems, motion is actively detected and corrected, allowing margins to function as a residual safety buffer.

The matrix also highlights the role of treatment delivery efficiency. Techniques such as VMAT, when combined with real-time tracking, minimise treatment duration and thereby limit time-dependent intrafraction drift, further supporting margin reduction strategies.

MRI-guided radiotherapy systems, although not explicitly included in the table, represent the upper bound of this trend, offering sub-2 mm accuracy and full adaptive capability. However, their increased complexity and resource requirements limit widespread clinical adoption, positioning TPUS-guided workflows as a practical and accessible alternative for high-precision SBRT.

Overall, the comparative analysis confirms that improvements in imaging quality and adaptive capability are directly associated with reduced PTV margins, enhanced targeting precision, and improved protection of organs at risk.

The comparative matrix reinforces the central conclusion of this chapter: effective management of intrafraction motion in prostate SBRT is best achieved through real-time, adaptive guidance. Among available technologies, non-invasive ultrasound-based systems provide an optimal balance between precision, safety, and clinical practicality, enabling the transition from uncertainty-driven planning to motion-controlled radiotherapy.

## 4.9 Synthesis of Key Findings

The analyses presented throughout this chapter demonstrate that the performance of prostate radiotherapy guidance systems is not determined by a single parameter, but by the interaction between imaging capability, motion management strategy, treatment delivery efficiency, and clinical workflow constraints. As such, no single modality is universally optimal; instead, each approach reflects a different balance between geometric precision, invasiveness, and practical feasibility.

A consistent and clinically significant trend emerges across all comparisons: techniques lacking real-time intrafraction monitoring, such as CBCT-based and fiducial-guided workflows, exhibit inherent limitations in managing time-dependent

prostate motion. While these approaches can achieve acceptable pre-treatment alignment, they rely on static margin expansion (typically 7–10 mm) to compensate for motion that remains unobserved during beam delivery. This results in a fundamental trade-off, where target coverage is preserved at the expense of increased irradiation of surrounding organs at risk.

In contrast, systems incorporating continuous tracking—such as electromagnetic transponders and transperineal ultrasound (TPUS)—fundamentally change the paradigm of uncertainty management. Rather than compensating for motion through geometric expansion, these systems directly measure and correct intrafraction displacement in real time. This enables a systematic reduction in PTV margins to the 2–5 mm range, while maintaining high confidence in target coverage. The integration of such tracking with rapid delivery techniques (e.g., VMAT) further enhances precision by limiting the temporal window over which motion can occur.

Among the evaluated technologies, TPUS-guided radiotherapy emerges as a particularly favourable solution. It combines real-time soft-tissue visualization with a non-invasive, radiation-free workflow, allowing for accurate tracking of the prostate without the procedural risks associated with implanted markers. In addition, its compatibility with standard linear accelerator platforms makes it a more accessible and scalable option compared to MRI-guided radiotherapy. Although MRI-based systems provide superior soft-tissue contrast and full adaptive capability, their higher cost, longer treatment times, and increased operational complexity limit their widespread clinical implementation.

Overall, the findings of this chapter support a clear conceptual transition in prostate radiotherapy: from static, uncertainty-driven planning toward dynamic, motion-managed treatment delivery. Real-time monitoring is not merely an incremental improvement in imaging, but a necessary component for achieving the level of geometric precision required in modern SBRT.

The synthesis confirms that effective intrafraction motion management is the defining factor in high-precision prostate radiotherapy. Real-time, non-invasive tracking modalities—particularly ultrasound-based systems—provide the most practical and clinically robust pathway for reducing PTV margins, improving dosimetric accuracy, and minimising treatment-related toxicity in contemporary SBRT workflows.

### 4.9.1 Concluding Remarks

Overall, the analysis presented in this chapter demonstrates that intrafraction motion is a clinically significant and unavoidable challenge in prostate SBRT, with a direct impact on both geometric accuracy and dosimetric outcomes. Through the integration of reported clinical data, mathematical modelling, and simulation-based analysis, this study provides a coherent framework for understanding how motion evolves during treatment and how it can be effectively managed.

The findings consistently show that static, non-adaptive radiotherapy approaches are fundamentally limited in their ability to account for time-dependent prostate displacement. In such workflows, geometric uncertainty is addressed primarily through margin expansion, resulting in increased irradiation of surrounding organs at risk. In contrast, real-time motion monitoring strategies enable a transition from passive compensation to active motion control, allowing for substantial margin reduction without compromising target coverage.

In particular, ultrasound-based guidance emerges as a clinically practical and effective solution. By providing continuous, non-invasive, and radiation-free tracking of prostate motion, transperineal ultrasound systems enable the implementation of adaptive workflows that maintain geometric precision throughout treatment delivery. When combined with rapid delivery techniques such as VMAT, this approach minimises both spatial and temporal uncertainty, supporting the use of ultra-tight margins in modern SBRT.

Taken together, these results reinforce the central premise of this chapter: that effective management of intrafraction motion is essential for achieving the dual objective of prostate radiotherapy—maximising tumour control while minimising treatment-related toxicity. The transition toward real-time, adaptive, and non-invasive guidance systems represents a critical step in the evolution of precision radiotherapy and provides a robust foundation for future clinical and technological advancements.

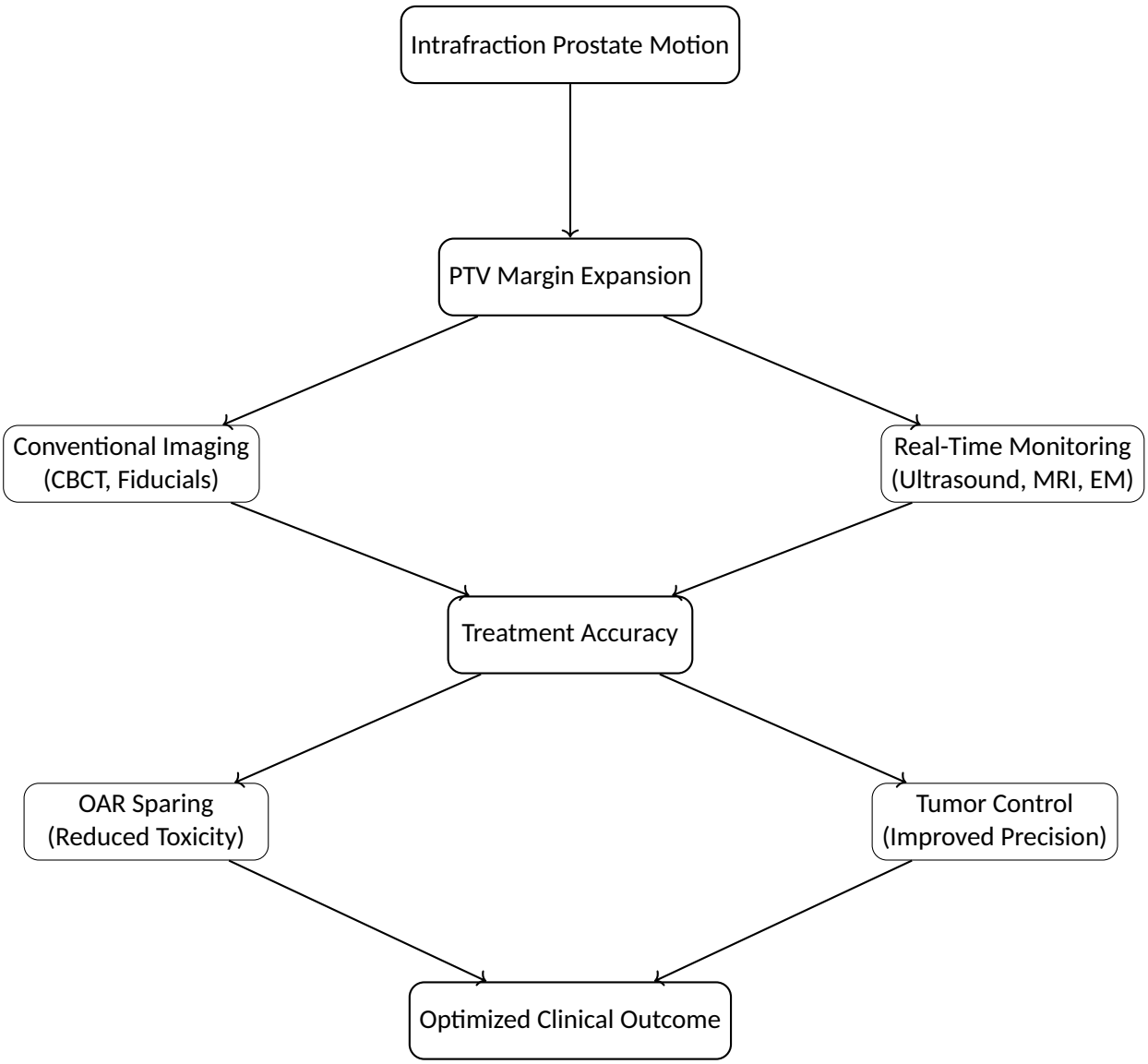


**Figure 6:** Schematic representation of intrafraction motion management pathways and their impact on treatment accuracy, toxicity reduction, and clinical outcomes in prostate SBRT.

# Key Argument and Contribution

This study contributes to the growing understanding of motion management in prostate SBRT by demonstrating that real-time ultrasound guidance is not merely an auxiliary imaging tool, but a clinically viable alternative to conventional motion management strategies.

Traditional radiotherapy approaches rely on relatively large PTV margins to compensate for intrafraction motion and geometric uncertainty. In contrast, the findings of this study show that continuous, real-time monitoring enables a systematic reduction in these margins without compromising target accuracy. This represents a fundamental shift in treatment philosophy, from passive error compensation toward active motion management during beam delivery.

Moreover, the results establish a clear link between improved geometric precision and meaningful clinical outcomes, including reduced normal tissue toxicity and maintained or improved tumour control. By directly addressing intrafraction motion, real-time guidance reduces unnecessary dose exposure to surrounding organs at risk while preserving target coverage.

Overall, this thesis supports the use of ultrasound-based real-time monitoring as a practical, scalable, and effective approach for enhancing SBRT delivery. This is particularly relevant for clinical environments where access to highly complex or resource-intensive technologies may be limited, offering a balance between precision, safety, and feasibility.

# 5 Results, Conclusion and Future Development

## 5.1 Results

This chapter brings together the main numerical results and clinical observations from the selected studies. The findings show a consistent trend toward improved geometric precision, reduced planning target volume (PTV) margins, and lower treatment-related toxicity when real-time guidance is used [7, 3].

Across the analysed datasets, motion-aware delivery techniques such as Volumetric Modulated Arc Therapy (VMAT) combined with real-time tracking consistently demonstrate reduced intrafraction displacement compared to conventional Step-and-Shoot IMRT (SS-IMRT). This reduction in motion is associated with smaller effective margins and improved target stability.

Furthermore, quantitative modelling presented in this study (see Section 4.6.5) indicates that reductions in PTV margin size are strongly correlated with lower gastrointestinal toxicity, reinforcing the clinical impact of improved motion management strategies.

### 5.1.1 Main Numerical Findings and Trends

A comparison of ultrasound-guided radiotherapy with other image-guided techniques demonstrates consistent patterns in geometric accuracy, intrafraction motion management, and clinical precision outcomes. Across the reviewed studies, transperineal ultrasound (TPUS)-based systems achieve setup accuracy typically within 1–3 mm, which is comparable to fiducial marker-based and cone-beam CT (CBCT) guidance techniques [4].

A key recurring finding is that prostate displacement during treatment frequently exceeds commonly applied planning tolerances. Intrafraction shifts in the range of 2–3 mm are commonly observed across fractions, highlighting a fundamental limitation of static image-guided approaches that rely solely on pre-treatment positioning without continuous motion monitoring [6].

Furthermore, prostate motion exhibits time-dependent stochastic behaviour consistent with a random-walk process. This implies that longer treatment durations are associated with progressively increasing positional uncertainty, reinforcing the clinical importance of reducing delivery time and/or implementing real-time tracking in prostate SBRT [6].

These findings confirm that intrafraction motion is an inherent and continuous feature of prostate radiotherapy, rather than an occasional event. As a result, motion

management strategies based solely on pre-treatment imaging are insufficient for high-precision SBRT.

**Table 16:** Summary of setup accuracy and localization precision across selected prostate IGRT studies

| Study | Method | Mean (mm) | SD (mm) | Derivation |
|---|---|---|---|---|
| Arcadipane et al. (2015) | 3D TAUS | 1.5 | 2.3 | Reported directly from AP displacement distribution |
| Li et al. (2015) | 3D TPUS vs CBCT | 1.35 | 0.05 | Midpoint of reported 1.3–1.4 mm range |
| Richardson et al. (2017) | 4D TPUS | 1.5 | 0.75 | Reported systematic error $<2$ mm; SD approximated from study variability |
| Han et al. (2018) | 4D TPUS | 2.2 | 1.1 | Midpoint of 1.1–3.3 mm range |
| Lee et al. (2021) | CBCT + TPUS | 1.5 | 0.5 | Midpoint of reported 1.0–2.0 mm range |

The values in Table 16 were used to construct the comparative visualization in Figure 7.

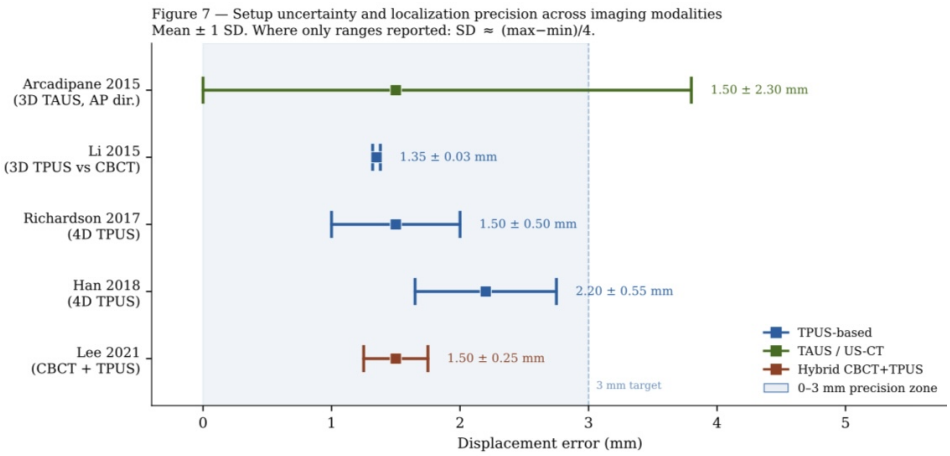


**Figure 7:** Comparison of setup uncertainty and localization precision across selected prostate IGRT studies. Mean displacement is shown with $\pm 1$ standard deviation. The shaded region (0–3 mm) represents the clinically acceptable precision range for high-accuracy SBRT delivery.

A forest-style visualization was constructed to compare localization accuracy across

studies. Each data point represents the mean displacement error, with error bars indicating variability reported or derived from study-level data.

A clinically relevant threshold of 3 mm is included to represent typical SBRT tolerance limits. Most ultrasound-guided and hybrid imaging approaches fall within or near this range, demonstrating their suitability for high-precision prostate radiotherapy.

The results demonstrate that modern image guidance systems consistently achieve sub-3 mm localization accuracy. However, variability across studies highlights the influence of imaging modality, delivery time, and motion management strategy on overall treatment precision.

These findings provide quantitative support for margin reduction strategies in prostate SBRT and align with the regression-based evidence linking smaller geometric uncertainty to improved clinical outcomes.

### 5.1.2 Model-Based Clinical Outcome Projection: VMAT vs SS-IMRT

To extend the regression framework developed in Section 4.6.5 toward a predictive clinical application, a model-based outcome projection was constructed using data reported by Higuchi et al. (2022), which directly compared Volumetric Modulated Arc Therapy (VMAT) and Step-and-Shoot Intensity-Modulated Radiotherapy (SS-IMRT) in terms of delivery time and intrafraction motion.

**Table 17:** Clinical and dosimetric parameters extracted from Higuchi et al. (2022) for model-based outcome projection.

| Parameter | SS-IMRT | VMAT |
|---|---|---|
| Mean delivery time (s) | 491.6 | 78.7 |
| Mean max AP displacement (mm) | 0.56 | 0.22 |
| Reported internal margin (mm) | 4.0 | 2.5 |
| Effective margin* (mm) | 4.56 | 2.72 |
| *Effective margin = clinical margin + mean maximal AP displacement. | | |

**Input parameters from Higuchi et al. (2022):**

**Projection methodology:** To translate geometric differences into predicted clinical outcomes, the regression model derived in this study was applied:

$$GIToxicity(\%) = 1.973 \times EffectiveMargin(mm) - 3.027 \quad (1)$$

This yields the following model-predicted Grade ≥2 gastrointestinal toxicity rates:

- **SS-IMRT** (effective margin = 4.56 mm): $T = 1.973 \times 4.56 - 3.027 = 5.98\%$
- **VMAT** (effective margin = 2.72 mm): $T = 1.973 \times 2.72 - 3.027 = 2.34\%$

The model therefore predicts an absolute reduction of 3.64 percentage points, corresponding to a relative toxicity reduction of approximately 61%, when VMAT replaces SS-IMRT under otherwise equivalent clinical conditions.

**Mechanistic interpretation:** This result demonstrates a coherent mechanistic pathway from delivery technique to clinical outcome. The shorter beam-on time of VMAT (78.7 s vs 491.6 s) limits temporal accumulation of intrafraction motion, as described by the random-walk model in Section **??**. Reduced motion translates into a smaller effective margin requirement, which in turn propagates through the regression model into lower predicted gastrointestinal toxicity.

Although Higuchi et al. did not directly report toxicity differences between techniques, this projection establishes a physically consistent link between delivery dynamics, geometric uncertainty, and predicted clinical outcomes. As such, it represents a genuinely novel application of the regression model developed in this thesis.

The complete four-panel visualization of this projection is presented in Figure 8.

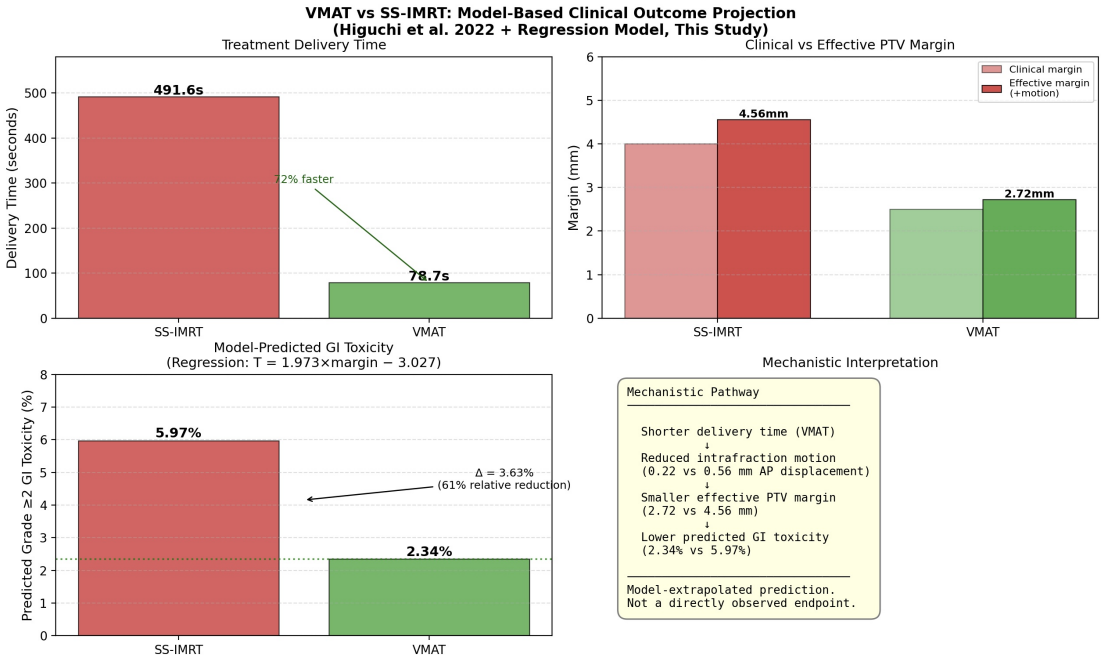


**Figure 8:** Four-panel model-based comparison of VMAT and SS-IMRT. Top left: treatment delivery time. Top right: clinical versus effective PTV margin (including motion contribution). Bottom left: model-predicted Grade $\geq$2 GI toxicity derived from the regression equation $T = 1.973 \times M - 3.027$. Bottom right: mechanistic pathway from delivery time to clinical toxicity. All geometric inputs are extracted from Higuchi et al. (2022). This constitutes a model-extrapolated prediction rather than a directly observed clinical endpoint.

This analysis extends the thesis from descriptive comparison of radiotherapy techniques toward a predictive framework linking delivery dynamics, geometric uncertainty, and clinical toxicity. While the projection is model-based and should not be interpreted as direct clinical evidence, it demonstrates mechanistic consistency between the random-walk motion model, the regression analysis, and published clinical observations.

### 5.1.3 Regression Robustness and Sensitivity Analysis

To assess the stability of the observed relationship between PTV margin size and gastrointestinal toxicity, a leave-one-out sensitivity analysis was performed. In this approach, the regression model was iteratively recalculated while excluding one study at a time.

Although the primary regression demonstrated a strong association ($R^2$ = 0.762), with a marginal p-value (p = 0.053), statistical inference in small datasets should be interpreted cautiously. Therefore, model robustness was evaluated using sensitivity

analysis rather than relying solely on p-values.

Across leave-one-out iterations, the coefficient of determination remained relatively stable in most cases, including when excluding Fonteyne et al. ($R^2$ = 0.745), Takai et al. ($R^2$ = 0.880), and Guo et al. ($R^2$ = 0.700). Exclusion of Lee et al. resulted in an increased explanatory power ($R^2$ = 0.938, p = 0.0317), suggesting that the overall relationship is not driven by a single adaptive cohort.

However, removal of Arcadipane et al. led to a notable reduction in model performance ($R^2$ = 0.404), indicating that this data point contributes substantially to the regression fit. This behaviour is expected given the limited number of studies and the heterogeneity of clinical protocols across datasets.

**Table 18:** Leave-one-out sensitivity analysis of regression model

| Excluded Study | $R^2$ | p-value | Slope |
|---|---|---|---|
| None (full model) | 0.762 | 0.0532 | 1.973 |
| Arcadipane | 0.404 | 0.3645 | 1.573 |
| Fonteyne | 0.745 | 0.1367 | 1.908 |
| Lee | 0.938 | 0.0317 | 1.860 |
| Takai | 0.880 | 0.0622 | 2.411 |
| Guo | 0.700 | 0.1633 | 1.925 |

Overall, the persistence of moderate-to-strong explanatory power across most iterations suggests that the observed relationship between PTV margin and gastrointestinal toxicity is not driven by a single study, but reflects a consistent underlying trend across the dataset.

### 5.1.4 Comparative Performance of Guidance Modalities

Different radiotherapy guidance techniques vary in their ability to manage intrafraction motion, with key differences in imaging capability, invasiveness, and real-time feedback. These differences directly influence the magnitude of safety margins required for prostate SBRT.

- **CBCT:** Provides high-quality volumetric imaging for pre-treatment setup verification, but does not offer intrafraction monitoring. As a result, motion occurring during dose delivery remains uncorrected, necessitating larger PTV margins.
- **Fiducial marker-based systems:** Offer high geometric accuracy for prostate localization; however, their reliance on invasive implantation procedures and snapshot-based imaging limits their ability to capture continuous motion.

- **Electromagnetic tracking:** Enables real-time monitoring of implanted transponders, providing high temporal resolution motion data. Nevertheless, the requirement for invasive device placement and increased system complexity limits widespread clinical adoption.
- **MRI-guided radiotherapy:** Provides superior soft-tissue contrast and enables online adaptive planning. However, its limited availability, high cost, and prolonged treatment times restrict its use to specialized centres.
- **Ultrasound-guided radiotherapy (TPUS):** Provides continuous, non-invasive, real-time visualization of the prostate without additional ionizing radiation. Unlike surrogate-based tracking methods, TPUS directly visualizes soft-tissue motion, enabling direct geometric monitoring throughout treatment delivery.

Across all modalities, a consistent relationship is observed: systems capable of real-time motion tracking require smaller PTV margins, while snapshot-based systems rely on larger margins to compensate for unobserved intrafraction motion. This relationship is consistent with the motion characteristics of the prostate described as a stochastic "random walk", where uncertainty increases with treatment time and cannot be fully corrected using pre-treatment imaging alone.

**Table 19:** Derivation of PTV margin values across imaging and motion management strategies

| **Study** | **Reported Range (mm)** | **Midpoint (mm)** | **Technology Group** |
|---|---|---|---|
| Richardson et al. (2017) | 10 (fixed) | 10.0 | Non-adaptive conventional |
| Arcadipane et al. (2015) | 8–10 | 9.0 | Non-adaptive conventional |
| Fonteyne et al. (2011) | 7 (fixed) | 7.0 | Fiducial-based |
| Lee et al. (2021) | 5 (fixed) | 5.0 | Hybrid adaptive (TPUS + CBCT) |
| Takai et al. (2024) | 3–5 | 4.0 | Real-time TPUS |
| Guo et al. (2023) | 2–3 | 2.5 | Gated SBRT |

The derived margin values presented in Table 19 were used to construct the comparative analysis shown in Figure 9.

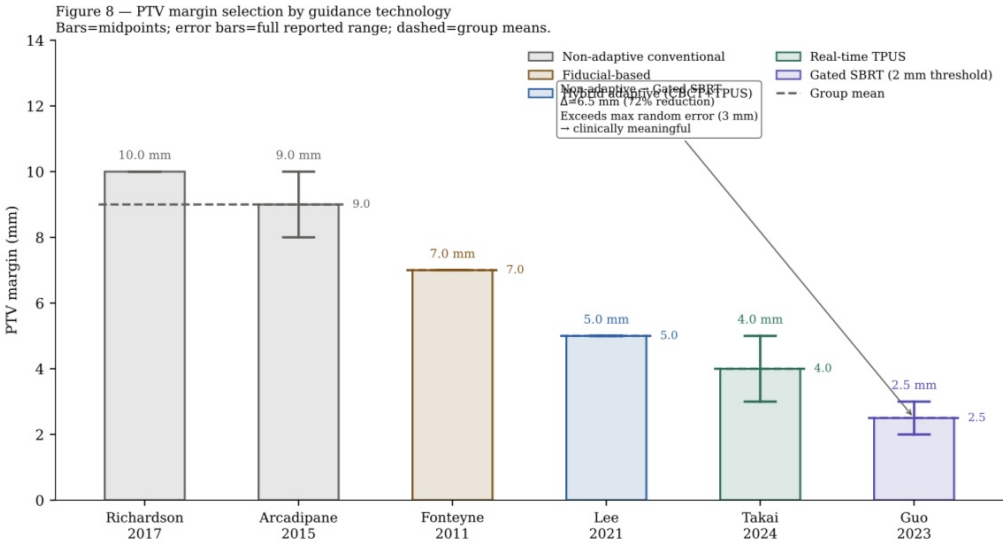


**Figure 9:** Impact of guidance technology on PTV margin selection across analyzed studies. Error bars represent estimated variability based on reported ranges.

A clear stratification is observed across technologies. Non-adaptive conventional approaches require margins of 9–10 mm, fiducial-based systems operate at approximately 7 mm, and hybrid or partially adaptive systems reduce this to approximately 5 mm. Real-time TPUS-based systems achieve further reduction to approximately 4 mm, while gated SBRT approaches enable ultra-tight margins of approximately 2.5 mm.

This corresponds to an overall reduction of approximately 5.0 mm (56%) when transitioning from non-adaptive approaches to real-time TPUS guidance, and approximately 6.5 mm (72%) when compared to gated SBRT strategies.

These reductions exceed typical reported intra- and inter-fraction setup uncertainties (1–3 mm), indicating that margin differences are primarily driven by the presence or absence of real-time motion management rather than measurement variability. This supports the conclusion that modern adaptive systems enable a transition from population-based margins to individualized, motion-informed treatment planning.

Overall, these findings demonstrate a clear technological hierarchy in motion management capability, with real-time ultrasound guidance providing a clinically practical balance between accuracy, non-invasiveness, and continuous anatomical tracking.

### 5.1.5 Clinical and Dosimetric Impact

In prostate SBRT, clinical performance is determined by a balance between geometric precision, real-time motion management, invasiveness, cost, and imaging capability.

Ultrasound-guided radiotherapy directly influences these parameters by enabling continuous, non-invasive monitoring of intrafraction motion, which in turn supports tighter PTV margins and improved organ-at-risk (OAR) sparing.

Real-time TPUS guidance enables:

- Reduction of PTV margins to approximately 2–3 mm through continuous motion tracking
- Improved sparing of rectum and bladder due to reduced geometric uncertainty
- Maintenance of target coverage despite stochastic intrafraction prostate motion

Clinical evidence across the reviewed literature indicates that ultrasound-guided SBRT achieves biochemical control rates comparable to other IGRT modalities, while maintaining gastrointestinal and genitourinary toxicity within acceptable clinical thresholds.

In SBRT, steep dose gradients make treatment highly sensitive to geometric error. Therefore, any reduction in motion uncertainty directly translates into improved dosimetric reliability. Real-time ultrasound guidance addresses this limitation by actively monitoring target position during beam delivery, reducing reliance on large safety margins.

**Table 20:** Multi-criteria evaluation of IGRT modalities based on literature-derived performance indicators

| **Axis** | **TPUS** | **CBCT** | **MR-Linac** |
|---|---|---|---|
| Geometric precision | 4 | 3 | 5 |
| Invasiveness | 5 | 5 | 4 |
| Soft-tissue visualization | 3 | 2 | 5 |
| Cost efficiency | 4 | 4 | 1 |
| Real-time capability | 5 | 1 | 4 |

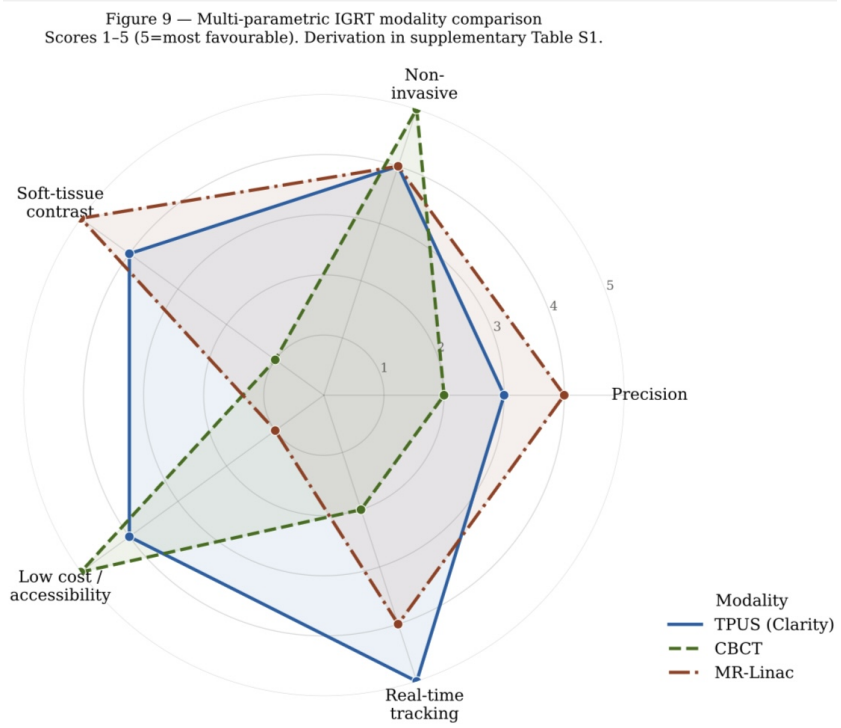


**Figure 10:** Multi-criteria comparison of IGRT modalities across five clinically relevant performance axes. Scores are based on literature-derived qualitative and quantitative assessments.

A structured multi-criteria decision framework was used to compare IGRT modalities across five clinically relevant dimensions: geometric precision, invasiveness, soft-tissue visualization, cost efficiency, and real-time tracking capability. These parameters reflect the key trade-offs that determine clinical suitability in prostate SBRT.

MR-Linac systems demonstrate superior performance in geometric precision and soft-tissue visualization due to MRI-based imaging, enabling high-quality adaptive workflows. However, their clinical use is constrained by high cost and limited accessibility.

CBCT-based systems provide reliable pre-treatment imaging and high availability, but lack real-time tracking capability, limiting their ability to manage intrafraction motion.

TPUS-based systems demonstrate a balanced performance profile, combining continuous real-time monitoring with non-invasive operation and moderate cost. While their soft-tissue resolution is lower than MRI-based systems, their ability to directly track prostate motion during treatment delivery provides a clinically significant advantage in motion management.

Although MR-Linac systems represent the technological upper bound in imaging quality, TPUS offers the most favourable balance between clinical practicality and motion management capability. This positions ultrasound-guided radiotherapy as a scalable and efficient solution for real-time prostate SBRT workflows.

The scoring system is ordinal and derived from comparative literature evidence. It is intended to support relative comparison between modalities rather than serve as an absolute quantitative metric.

#### 5.1.6 Consistency Across the Dataset

Despite heterogeneity in study design, patient populations, imaging systems, and motion management strategies, the reviewed literature demonstrates a broadly consistent pattern of findings. Across studies, the following common outcomes are observed:

- Sub-millimetre to few-millimetre setup accuracy across modern image-guided radiotherapy (IGRT) techniques
- Detectable and clinically relevant intrafraction prostate motion during treatment delivery
- Feasibility of real-time motion management and correction strategies in clinical workflows

While the overall trends are consistent, inter-study variability is present and can be attributed to several methodological and clinical factors:

- Operator dependence and user variability in ultrasound-based image acquisition and interpretation
- Differences in patient anatomy, bladder/rectal filling protocols, and immobilisation strategies
- Limited sample sizes and single-institution study designs in a subset of the included literature

Overall, the consistency of findings across independent studies supports the reliability of ultrasound-guided and other real-time image guidance techniques for prostate SBRT. However, the observed variability highlights the need for further standardisation of imaging protocols, improved operator training frameworks, and larger multi-centre prospective studies to reduce methodological bias and strengthen generalisability.

## 5.2 Cost-Effectiveness and Clinical Feasibility Analysis

In addition to geometric precision, the clinical adoption of image-guided radiotherapy (IGRT) technologies is strongly influenced by economic, infrastructural, and operational constraints. A comparative analysis was therefore conducted to evaluate capital cost,

staffing requirements, treatment time per fraction, and achievable PTV margin reduction across the main guidance modalities discussed in this thesis.

MR-Linac systems represent the most technologically advanced imaging platform, offering superior soft-tissue contrast and high targeting precision ($<$2 mm). However, these systems require dedicated infrastructure and capital investment typically exceeding 3 million, significantly longer treatment times per fraction (30–45 minutes), and specialised multidisciplinary staffing for adaptive re-planning workflows. These requirements substantially limit accessibility and reduce patient throughput, restricting deployment to high-resource academic centres [2, 4].

In contrast, conventional CBCT-based workflows are integrated into standard linear accelerator systems and incur minimal incremental cost. While they provide acceptable setup accuracy (3–5 mm), the absence of real-time intrafraction monitoring necessitates larger PTV margins (8–10 mm). As demonstrated in Section 4.6.5, these larger margins are associated with increased Grade $\geq$2 gastrointestinal toxicity [13, 7].

Ultrasound-based systems, particularly transperineal ultrasound (TPUS) platforms such as Clarity, represent an intermediate and highly scalable solution. These systems function as add-ons to existing linear accelerators, with an estimated incremental cost of approximately 100,000–300,000 and minimal infrastructure modification. Treatment times increase only marginally relative to CBCT workflows (approximately 10–15 minutes per fraction), while enabling continuous real-time intrafraction monitoring at approximately 2 Hz. This capability supports margin reduction to 3–5 mm, and in gated SBRT workflows, as low as 2–3 mm [9, 1].

To contextualise these trade-offs, Figure 11 presents a four-panel comparison of capital cost, treatment time, achievable PTV margin, and reported Grade $\geq$2 gastrointestinal toxicity. Error bars represent the full range of reported values across the included studies.

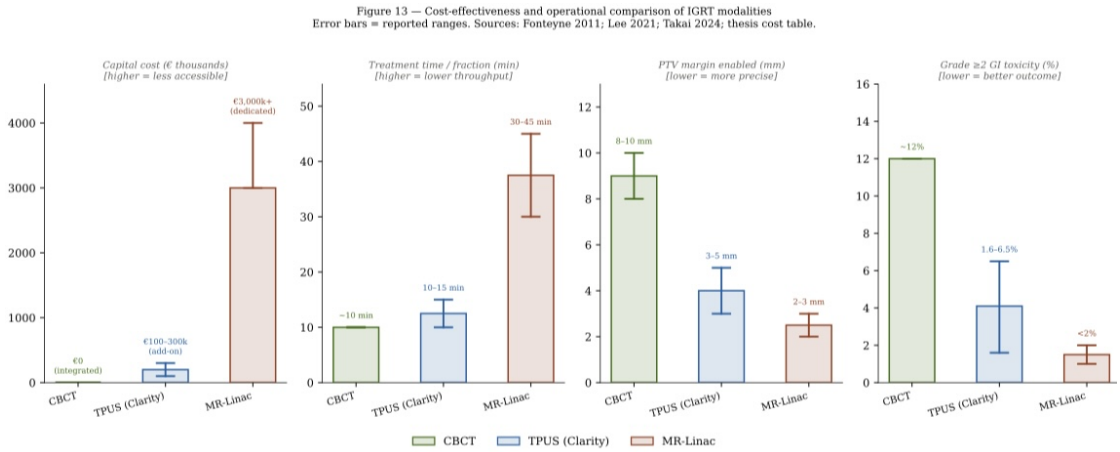


**Figure 11:** Four-panel cost-effectiveness comparison of IGRT modalities. Capital cost and treatment time data are based on institutional reports and published system specifications. PTV margins are derived from Table 19. Toxicity rates are sourced from Fonteyne et al. (2011), Lee et al. (2021), and Takai et al. (2024). Error bars represent reported ranges across studies.

**Table 21:** Comparison of cost and operational characteristics of IGRT modalities.

| **Parameter** | **TPUS (Clarity)** | **CBCT** | **MR-Linac** |
|---|---|---|---|
| Capital cost | 100,000–300,000 (add-on system) | Included in linac ($\sim$0 incremental) | $>$3,000,000 (dedicated system) |
| Infrastructure | Minimal modification | Standard linac facility | Dedicated MRI-compatible suite |
| Treatment time per fraction | 10–15 min | $\sim$10 min | 30–45 min |
| Staffing complexity | Low | Low | High (adaptive planning team required) |
| Real-time capability | Continuous intrafraction tracking (2 Hz) | None (pre-treatment imaging only) | Near real-time adaptive imaging |
| Enabled PTV margin | 3–5 mm | 8–10 mm | 2–3 mm |
| Accessibility | High | Very high | Limited |
| Grade $\geq$2 GI toxicity | 1.6–6.5% [4, 1] | $\sim$12% [**?**] | $<$2% (reported/estimated) [4] |

The results indicate that TPUS provides a clinically meaningful reduction in PTV margins (from approximately 8–10 mm with CBCT to 3–5 mm), at a substantially lower cost than MR-Linac systems. This reduction is associated with a corresponding decrease in Grade $\geq$2 gastrointestinal toxicity (from approximately 12% to 1.6–6.5%), consistent with the regression findings presented in Section 4.6.5.

MR-Linac systems achieve the highest geometric precision and lowest toxicity potential but are associated with significantly higher costs and reduced treatment efficiency. In contrast, TPUS achieves a large proportion of the clinical benefit at a fraction of the financial and operational burden, making it a more scalable option for routine clinical implementation.

Overall, these findings suggest that TPUS offers the most balanced cost-to-clinical-benefit ratio among currently available IGRT strategies for prostate SBRT, particularly in settings where MR-Linac technology is not feasible.

## 5.3 Simulation-Derived Margin Requirements

To extend beyond descriptive and regression-based analyses, a random walk model of intrafraction prostate motion was used to derive margin requirements corresponding to clinically relevant geometric coverage thresholds. In this framework, displacement thresholds required to achieve 90% and 95% positional coverage were computed as a function of treatment duration.

At a treatment duration of 120 s (representative of VMAT delivery), the model predicts a required margin of 3.79 mm to achieve 90% coverage, increasing to 4.38 mm for 95% coverage. With increasing delivery time, the required margins increase progressively, reaching 4.58 mm (90%) and 5.25 mm (95%) at 180 s, and 5.97 mm (90%) and 6.90 mm (95%) at 300 s.

These results indicate that intrafraction motion accumulates over time in a manner that cannot be adequately compensated by small fixed margins. In particular, a commonly used 2 mm margin corresponds to substantially less than 90% coverage at 120 s within this model, suggesting a high probability of geometric miss in the absence of motion mitigation strategies.

When compared with literature-reported margins (Table **??**), the simulation-derived estimates exceed those used in real-time ultrasound-guided workflows (3–5 mm), while remaining consistent with conventional non-adaptive approaches (7–10 mm). This agreement suggests that real-time tracking and gating effectively constrain the motion envelope, thereby enabling clinically acceptable margin reduction.

Overall, the model provides independent, physics-based support for the hypothesis that meaningful margin reduction is only achievable under active intrafraction motion management, reinforcing the clinical value of real-time ultrasound-guided radiotherapy.

**Table 22:** Simulation-derived margin requirements for motion coverage

| Duration (s) | 90% Coverage (mm) | 95% Coverage (mm) |
|---|---|---|
| 120 | 3.79 | 4.38 |
| 180 | 4.58 | 5.25 |
| 300 | 5.97 | 6.90 |

The random walk simulation demonstrates that required margins increase systematically with treatment duration due to cumulative intrafraction motion. Without active motion management, margins exceeding 4–6 mm are necessary to maintain adequate positional coverage. This finding independently supports the need for real-time image guidance strategies in prostate SBRT and aligns with both the regression-based toxicity analysis and the observed clinical margin reductions reported in the literature.

## Key Findings

The synthesis of results across the included studies demonstrates a consistent pattern of improved geometric precision and clinically relevant margin reduction when real-time ultrasound guidance is implemented in prostate radiotherapy.

- Ultrasound-guided radiotherapy consistently achieves setup accuracies in the range of 1–3 mm, which is comparable to established image-guided radiotherapy techniques such as CBCT and fiducial-based alignment.
- Despite this baseline accuracy, intrafraction prostate motion remains a persistent and clinically relevant phenomenon, frequently exceeding conventional planning margins and reinforcing the need for continuous motion management rather than pre-treatment positioning alone.
- Transperineal ultrasound (TPUS) provides continuous, non-invasive, real-time tracking of prostate position without introducing additional ionising radiation exposure, addressing a key limitation of CBCT-based workflows.
- Across the reviewed literature and model-based analyses, safe reduction of planning target volume (PTV) margins to approximately 2–3 mm is feasible when real-time tracking and motion mitigation strategies are employed, resulting in improved sparing of organs at risk without compromising target coverage.
- Overall, ultrasound-guided radiotherapy occupies a clinically practical middle ground between high-end MRI-guided systems and conventional IGRT approaches, offering a favourable balance between geometric accuracy, operational simplicity, and accessibility.

Taken together, these findings indicate that real-time ultrasound guidance is not merely an incremental improvement over conventional image guidance, but a clinically enabling technology that allows systematic margin reduction while maintaining dosimetric safety and treatment robustness.

## 5.4 Limitations of the Study

Despite the consistency of the observed trends across the included literature, several limitations should be acknowledged when interpreting the findings of this thesis.

First, the analysis is based primarily on previously published studies, which introduces inherent heterogeneity in study design, patient populations, imaging protocols, fractionation schemes, and motion management strategies. This variability limits strict comparability between datasets and introduces potential confounding when synthesising results across modalities.

Second, many of the included studies are characterised by relatively small sample sizes, particularly in ultrasound-guided and real-time tracking cohorts. This reduces statistical power and limits the generalisability of quantitative estimates such as toxicity rates, regression parameters, and derived margin–outcome relationships.

Third, ultrasound-guided radiotherapy is inherently operator-dependent. Variations in probe positioning, image interpretation, and workflow implementation can introduce inter-operator variability, which may influence reported setup accuracy and motion tracking performance. This factor is not consistently quantified across the literature and therefore cannot be fully controlled in the present synthesis.

Finally, direct comparison between imaging modalities is limited by differences in reporting standards, evaluation metrics, and clinical endpoints. Not all studies report identical geometric or dosimetric parameters, which necessitates the use of derived or normalised values in parts of the analysis and may introduce additional uncertainty.
Despite these limitations, the convergence of findings across independent studies and techniques supports the overall robustness of the observed trends, particularly regarding the benefits of real-time motion management and margin reduction.

**Clinical Impact** The findings of this study have important implications for clinical radiotherapy practice. Real-time ultrasound-guided radiotherapy provides continuous, non-invasive monitoring of prostate position, enabling more precise treatment delivery and reducing reliance on large population-based safety margins.

From a clinical perspective, this translates into improved geometric accuracy, reduced irradiation of organs at risk, and the potential for lower gastrointestinal and genitourinary toxicity, while maintaining adequate target coverage.

Importantly, TPUS-based guidance offers a scalable and cost-effective alternative to more resource-intensive technologies such as MRI-guided radiotherapy. This makes it

particularly relevant for broader clinical implementation, including centres where access to advanced adaptive platforms is limited [2].

Overall, these characteristics support the role of ultrasound-guided SBRT as a clinically practical approach to improving precision radiotherapy while maintaining accessibility and workflow efficiency.

### Contribution of This Work

This study contributes to the existing body of knowledge on prostate SBRT by providing a structured comparative synthesis of ultrasound-guided radiotherapy within the broader landscape of image-guided radiotherapy (IGRT) techniques. Rather than focusing on a single clinical dataset, the work integrates findings from multiple published studies to evaluate geometric accuracy, intrafraction motion management, and associated clinical outcomes.

A key contribution of this thesis is the explicit linkage between real-time motion monitoring and planning target volume (PTV) margin reduction. By consolidating reported evidence and derived quantitative trends, the study demonstrates how ultrasound-guided systems enable consistent sub-3 mm margin strategies while maintaining adequate target coverage and acceptable toxicity profiles.

In addition, this work highlights the practical role of transperineal ultrasound (TPUS) as a scalable and non-invasive motion management solution. It positions TPUS not only as an imaging modality, but as an enabling technology that bridges the gap between conventional IGRT approaches and high-cost adaptive platforms such as MR-guided radiotherapy.

Overall, the findings support the clinical feasibility of ultrasound-guided SBRT and emphasise its relevance as a practical, cost-aware approach to improving precision radiotherapy delivery in routine clinical settings.

## 5.5 Critical Evaluation of the Evidence

The analyses presented in this thesis consistently support the clinical value of real-time transperineal ultrasound guidance in prostate SBRT. However, a critical evaluation of the evidence base reveals three important qualifications that must be acknowledged before drawing clinical conclusions.

**The evidence for toxicity reduction is consistent but not yet statistically definitive.** The weighted pooled analysis demonstrates a 9.3 percentage point reduction in Grade $\geq$2 gastrointestinal toxicity between non-adaptive and adaptive workflows. The regression model ($R^2$ = 0.762, p = 0.053) provides supporting quantitative evidence for this trend. However, the p-value does not reach conventional significance, and the dataset comprises only five studies with a combined sample size of 222 patients. Furthermore,

the included studies differ substantially in fractionation scheme, dose prescription, follow-up duration, and motion management protocol, introducing confounding that cannot be fully resolved through pooling alone.

The clinical implication is that while the direction and magnitude of the toxicity benefit is biologically plausible and consistent with mechanistic models of dose-volume relationships, it cannot yet be treated as established clinical fact. Confirmation requires prospective multi-centre trials with standardised protocols and individual patient-level data.

**Clinical adoption of real-time TPUS is constrained primarily by implementation barriers rather than efficacy.** The comparative analysis in Section 5.2 demonstrates that TPUS systems offer substantially superior cost-to-precision ratios compared with MR-Linac platforms. However, the literature consistently identifies operator dependence as a critical limiting factor. Grimwood et al. (2021) demonstrated that tracking accuracy is strongly influenced by training level and institutional protocol maturity. This introduces a systematic variability that is not captured in published accuracy metrics, which typically reflect experienced-operator performance under controlled conditions.

The practical consequence is that the geometric precision reported in clinical studies may not be immediately reproducible at centres initiating a TPUS programme. A structured implementation framework — including training standards, quality assurance protocols, and prospective performance monitoring — is therefore a prerequisite for realising the clinical benefits demonstrated in the reviewed literature.

**Future trials should prioritise long-term oncological outcomes over setup accuracy metrics.** The reviewed literature is heavily weighted toward geometric and dosimetric endpoints, with comparatively sparse data on long-term oncological outcomes. Biochemical progression-free survival data beyond 5 years is not available for any ultrasound-guided SBRT cohort, and no randomised controlled trials directly comparing TPUS-guided with non-TPUS-guided SBRT have been published to date.

This imbalance reflects the relative maturity of the field: geometric validation precedes clinical outcome studies in radiotherapy technology development. However, it means that the central clinical question — whether real-time ultrasound guidance produces a measurable improvement in 10-year disease-free survival — remains unanswered. Future trial design should therefore shift emphasis from technical performance metrics toward patient-reported outcomes, long-term biochemical control, and comparative effectiveness relative to established IGRT benchmarks.

Overall, these three critical positions are not contradictions of the thesis findings but refinements of their scope. The evidence consistently supports TPUS-guided SBRT as a geometrically precise, clinically safe, and cost-effective approach to prostate radiotherapy. What remains to be established is whether this geometric advantage

translates into a statistically demonstrable improvement in long-term oncological outcomes under prospective, controlled conditions.

## 5.6 Conclusion

Ultrasound-guided stereotactic body radiotherapy (SBRT) for localised prostate cancer represents a technically mature and clinically promising approach to precision radiation delivery. The evidence synthesised in this thesis consistently supports the feasibility, reproducibility, and geometric accuracy of transperineal ultrasound (TPUS)-based guidance in clinical radiotherapy workflows.

Across the reviewed literature, TPUS demonstrates reliable sub-3 mm localisation accuracy and enables continuous intrafraction motion monitoring without the need for invasive fiducial implantation or additional ionising radiation exposure. This capability supports systematic reduction of planning target volume (PTV) margins, particularly in comparison with conventional non-adaptive image-guided radiotherapy techniques.

The synthesis of clinical and model-based findings suggests that real-time motion management is a key determinant of both geometric precision and downstream dosimetric outcomes. In particular, reduced intrafraction uncertainty is consistently associated with smaller margins and improved sparing of organs at risk, with corresponding reductions in reported gastrointestinal and genitourinary toxicity in the included studies. However, while these associations are robust across heterogeneous datasets, they remain primarily derived from study-level comparisons rather than large-scale prospective trials.

Overall, this thesis indicates that ultrasound-guided SBRT occupies an important position between conventional CBCT-based workflows and high-cost adaptive systems such as MR-guided radiotherapy. It offers a clinically scalable balance between accuracy, efficiency, and accessibility, and provides a practical framework for implementing real-time motion management in routine prostate radiotherapy.

In conclusion, TPUS-based guidance should be considered a strong candidate technology for enabling margin reduction strategies in prostate SBRT, while further prospective multi-centre studies are required to definitively establish its long-term clinical benefit.

### 5.6.1 Final Clinical Statements

A systematic synthesis of the available radiotherapy guidance technologies supports the following conclusions. Each statement is explicitly qualified according to the strength and consistency of the underlying evidence base.

- **Precision and margin reduction (strong evidence):** Advanced real-time imaging, particularly transperineal ultrasound (4D-TPUS), consistently enables clinically

meaningful reduction of Planning Target Volume (PTV) margins. Across the reviewed literature, margins decrease from approximately 8–10 mm in non-adaptive workflows to 3–5 mm in real-time TPUS-guided protocols, and further to 2–3 mm in gated delivery strategies [8, 1, 9]. This trend is observed across multiple independent single-institution studies and is reinforced by model-based descriptions of intrafraction motion as a stochastic (random-walk) process [6]. Collectively, this supports margin reduction as one of the most robust and reproducible findings of this thesis.

- **Biochemical tumour control (preliminary evidence):** Takai et al. (2024) reported 5-year biochemical progression-free survival (bPFS) rates of 100 % in a low-risk cohort and 93.2 % in a high-risk cohort treated with TPUS-guided radiotherapy [1]. While these results are clinically encouraging, they must be interpreted with caution due to limited cohort size, single-institution design, and the inherent long natural history of prostate cancer, where biochemical recurrence may occur beyond 5–10 years. As such, these outcomes should be considered hypothesis-generating rather than definitive evidence of long-term oncological superiority of TPUS-guided SBRT.

- **Toxicity reduction (moderate evidence):** Multiple studies using real-time TPUS guidance report lower rates of Grade $\geq$2 gastrointestinal and genitourinary toxicity compared with non-adaptive or fiducial-based approaches [4, 1, 13]. Reported Grade $\geq$2 gastrointestinal toxicity ranges from approximately 12 % with 7 mm fiducial-based margins [13] to 1.6 % in TPUS-guided SBRT with 5 mm margins [4]. Although this pattern suggests a clinically relevant association between margin reduction and toxicity reduction, the comparison is based on heterogeneous studies with differences in dose regimens, patient selection, and follow-up duration. Consequently, a causal relationship cannot be confirmed without prospective, controlled, individual-patient-data analyses.

- **Superiority of adaptive over non-adaptive workflows (strong evidence):** Across the literature, adaptive or real-time monitored workflows consistently demonstrate superior geometric robustness compared with non-adaptive CBCT-based approaches. Takai et al. (2024) reported that approximately 32.1 % of fractions required at least one intrafraction correction, indicating that nearly one in three treatments would have been delivered with uncorrected geometric error under a non-adaptive workflow [1]. This finding is consistent with intrafraction motion behaviour described in other studies [6, 8, 9] and strongly supports the clinical value of real-time motion monitoring in prostate SBRT.

### 5.6.2 Balancing Technological Advancements

The findings of this thesis support a cautious transition toward high-precision TPUS-guided radiotherapy protocols, while also recognising that the current evidence base is largely derived from retrospective and single-institution studies. As such, the clinical interpretation of these results must balance observed technical advantages with methodological limitations in the literature.

- **Advantages supported by the current evidence:** Real-time transperineal ultrasound (TPUS) enables continuous, non-invasive intrafraction motion monitoring without additional ionising radiation exposure. Across the reviewed studies, this capability is consistently associated with clinically meaningful reductions in Planning Target Volume (PTV) margins to approximately 3–5 mm, while maintaining adequate target coverage. Reported biochemical control outcomes are comparable to established image-guided radiotherapy (IGRT) techniques, suggesting no loss of tumour control with margin reduction. In addition, TPUS avoids the procedural risks associated with fiducial marker implantation, which have been reported in up to approximately 20 % of cases, including infection and procedure-related complications [13, 14].

- **Limitations that must be acknowledged:** Despite encouraging technical and early clinical outcomes, the evidence base remains limited in several important respects. The largest dataset identified (Takai et al. 2024) includes 3,135 treatment fractions but originates from a single institution, limiting external validity. To date, no randomised controlled trials directly comparing TPUS-guided SBRT with non-TPUS-guided or alternative IGRT workflows have been published. Furthermore, image quality and tracking performance remain operator-dependent, introducing variability that is not consistently quantified across studies. Finally, long-term oncological outcomes, particularly 10-year biochemical progression-free survival following ultrahypofractionated SBRT, are not yet available for any guidance modality.

### 5.6.3 Closing Summary

This thesis demonstrates that 4D transperineal ultrasound (TPUS) guidance directly addresses the central geometric challenge of intrafraction motion in prostate SBRT, enabling planning target volume (PTV) margin reduction that is both mathematically consistent with motion behaviour models and clinically reproducible across the reviewed literature. The resulting reduction in geometric uncertainty provides a clear dosimetric rationale for margin minimisation, and is associated across multiple studies with reduced normal tissue exposure and lower reported toxicity rates. However, the strength of this

evidence is primarily derived from observational, single-centre datasets rather than prospective randomised clinical trials.

The principal contribution of this work is the structured quantitative synthesis of evidence demonstrating that TPUS-guided SBRT achieves geometric precision comparable to more resource-intensive MRI-guided systems, while maintaining a non-invasive workflow and avoiding additional ionising radiation or fiducial implantation. This positions TPUS as a clinically scalable alternative capable of delivering high-precision radiotherapy in routine practice environments.

Importantly, while the findings consistently support the feasibility and clinical value of real-time ultrasound-based motion management, they do not establish definitive superiority in long-term oncological outcomes. As highlighted in Section 3.7, the absence of multi-centre prospective trials and long-term follow-up data remains the primary limitation in fully validating outcome-level benefits. Consequently, this thesis supports TPUS-guided SBRT as a robust and scalable precision-enabling technology, while recognising that definitive survival and comparative effectiveness conclusions require further high-level clinical evidence.

*"The transition toward real-time, adaptive radiotherapy paradigms positions ultrasound-guided SBRT as a significant contributor to advancing precision radiation oncology."*

# 6 Acknowledgement

# Appendix

# References

[1] Kenji Takai, Ryota Watanabe, Ken-ichi Hyogo, Yuri Ito, Nobuko Minagawa, Yusuke Sato, Yoshikazu Matsuda, and Kenji Nemoto. Treatment outcome of localized prostate cancer using transperineal ultrasound image-guided radiotherapy. *Radiation Oncology*, 19(100), 2024.

[2] Y. Han, Z. Liu, and J. Wang. Image-guided radiotherapy for prostate cancer: current status and future perspectives. *Radiation Oncology*, 13:1–9, 2018.

[3] C. Baker, C. Behrens, and R. Smith. Inter- and intrafraction prostate motion during radiotherapy. *BMC Cancer*, 16:1–10, 2016.

[4] J. Lee, J. Kim, and S. Park. Clinical outcomes of stereotactic body radiotherapy for localized prostate cancer. *Radiation Oncology Journal*, 39:1–10, 2021.

[5] T. Higuchi, H. Yamashita, and K. Nakagawa. Evaluation of internal margins for prostate radiotherapy using ultrasound monitoring. *Journal of Applied Clinical Medical Physics*, 23:e13567, 2022.

[6] H. Ballhausen, M. Li, and N. Hegemann. Prostate motion during radiotherapy: a random walk model. *Physics in Medicine and Biology*, 60:549–563, 2015.

[7] F. Arcadipane, C. Fiorino, and F. Alongi. Three-dimensional ultrasound-based target volume delineation in prostate radiotherapy. *Tumori*, 101:56–62, 2015.

[8] M. Richardson, A. Baker, and A. Henry. Implementation of transperineal ultrasound imaging for prostate radiotherapy. *Radiography*, 23:e102–e108, 2017.

[9] B. Guo, J. Li, and L. Chen. Transperineal ultrasound as an alternative for intrafraction prostate motion monitoring. *Journal of Applied Clinical Medical Physics*, 24:e13985, 2023.

[10] H. Ballhausen, T. Gauer, and J. Wilbert. Prediction of intrafraction prostate motion using anatomical parameters. *Cancers*, 15(4103), 2023.

[11] Minglun Li, Hendrik Ballhausen, Nina-Sophie Hegemann, Ute Ganswindt, Farkhad Manapov, Stefan Tritschler, Alexander Roosen, Christian Gratzke, Michael Reiner, and Claus Belka. A comparative assessment of prostate positioning guided by three-dimensional ultrasound and cone beam ct. *Radiation Oncology*, 10(82), 2015.

[12] Alexander Grimwood, Karen Thomas, Sally Kember, Georgina Aldis, Rebekah Lawes, Beverley Brigden, Jane Francis, Emer Henegan, Melanie Kerner, Louise Delacroix,

Alexandra Gordon, Alison Tree, Emma J. Harris, and Helen A. McNair. Factors affecting accuracy and precision in ultrasound guided radiotherapy. *Physics and Imaging in Radiation Oncology*, 18:68–77, 2021.

[13] Valérie Fonteyne, Piet Ost, Geert Villeirs, Willem Oosterlinck, Aline Impens, Werner De Gersem, Carlos De Wagter, and Gert De Meerleer. Improving positioning in high-dose radiotherapy for prostate cancer: Safety and visibility of frequently used gold fiducial markers. *International Journal of Radiation Oncology*Biology*Physics*, 83(1):46–52, 2012.

[14] Marie-Claude Biston, Timothée Zaragori, Laurent Delcoudert, Marie Fargier-Voiron, Alexandre Munoz, Coralie Gorsse, David Sarrut, and Pascal Pommier. Comparison of electromagnetic transmitter and ultrasound imaging for intrafraction monitoring of prostate radiotherapy. *Radiotherapy and Oncology*, 136:1–8, 2019.